

WDW-3

Normal Deformation as Constrained Refoliation

Groupoid-faithful hypersurface-deformation closure under AASC

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Exact scope. The theorem closes the classical normal-deformation burden on the exact WDW-2 handoff. It proves the declared off-shell constraint algebra, occupied regular characteristic reduction, and guarded history-relative refoliation groupoid while retaining singular KID strata and standing-bearing boundary data. Quantum constraints, anomaly control, and physical-state claims remain outside scope.

Abstract

This paper closes the classical normal-deformation burden left open by the accepted WDW-1 canonical reconstruction and WDW-2 proper spatial-redescription quotient. The inherited object is retained in full: smooth regular cotangent carriers, a loss-aware Sobolev tower, the proper spatial action groupoid, structured source fibres and attachments, reconstructed histories, the complete equivariant lapse family, singular objects, and the boundary and large-transformation guards. On a declared algebra of smooth regular local functionals we calculate, off shell and in the inherited sign and density convention, the complete vacuum and minimally coupled massive Klein–Gordon hypersurface-deformation brackets. The normal–normal calculation is displayed term by term; field-dependent structure functions are retained in the off-shell function algebra rather than frozen as constants. The off-shell identity is not misidentified with a pointwise phase-space anchor: on the constraint surface the ideal quotient $\mathcal{I}/\mathcal{I}^2$ supplies the C^∞ -linear conormal Lie–Rinehart/algebroid representation.

For the vacuum branch, the full constraint map is proved split regular on an occupied no-Killing-initial-data chart on a fully typed Douglis–Nirenberg scale. Occupancy is explicit: on a closed hyperbolic three-manifold, the data $K = h$, $\Lambda = 0$, and $\pi^{ij} = -(\sqrt{h}/\kappa)h^{ij}$ satisfy the vacuum constraints; the KID equations reduce to the conformal-Killing equation, and negative curvature forces the lapse and shift components to vanish. The raw differential composition $A_z A_z^\#$ has unequal Douglis–Nirenberg row weights and is not inverted on the inherited equal-row target. A fixed, Sobolev-index-independent elliptic row balancer $\mathcal{B}_{z,*}$ instead defines $L_z = A_z \mathcal{B}_{z,*} A_z^\#$, a self-adjoint, index-zero, order-two elliptic pseudodifferential operator on that exact target. Its unique Green operator is the same equation inverse on every Sobolev level; uniform parameter estimates give $Q_z = \mathcal{B}_{z,*} A_z^\# L_z^{-1}$, a compatible smooth-tame right inverse, and a split zero locus. A single distributional-multiplier argument, followed by elliptic regularity, proves that the kernel of the pulled-back weak canonical form is exactly the distribution generated by all proper lapse-shift constraints. KID points are retained as singular groupoid objects with isotropy and second-order obstruction data. The Klein–Gordon off-shell algebra is unconditional; its full regular quotient is stated only where the independent ScalarReg3 certificate is populated, in accordance with the published scalar linearization-stability erratum.

Finite refoliation is not represented as a global Hamiltonian group action on instantaneous phase space. We construct instead a history-relative groupoid of admissible Cauchy embedding changes and a guarded canonical characteristic monodromy groupoid. After the sign-aligned on-shell morphism is proved with its coefficient terms, evaluation is defined directly on guarded path and homotopy classes; no infinite-dimensional Lie-II theorem is imported. On a shifted smooth Sobolev tower, the multiplier is retained in the target dual until elliptic regularity promotes it, and the equivalent formula $(d\text{Eval})^{-1} = -v_{\text{emb}} L_z^{-1} A_z \mathcal{B}_{z,*} \mathcal{J}_{E,z} \Omega^b$ is typed through the fixed normal operator. It satisfies uniform smooth-tame estimates. Hamilton’s Nash–Moser theorem then identifies embedding orbits with characteristic plaques; no-KID injectivity and trivial local isotropy and holonomy then derive the pair-groupoid form. Proper deformation routes with the same endpoints consequently determine the same canonical arrow. Global residuals are classified exhaustively as KID isotropy, large or mapping-class action, boundary charge, source or attachment change, relational record change, target change, or failure of existence or regularity. Proper refoliations preserve the structured source fibre, reconstruction ancestry, Dirac and relational profiles, and boundary certificate; they create no distinct terminal standing merely by changing slicing. Charge-bearing boundary evolution, large transformations, singular strata, and relationally distinct histories are not collapsed. The result is a source-fibre-preserving, boundary-standing-sensitive, groupoid-faithful classical refoliation closure and an exact branchwise handoff to WDW-4.

Keywords: AASC; ADM; canonical gravity; hypersurface-deformation algebra; refoliation; Lie algebroid; Lie groupoid; Killing initial data; characteristic reduction; boundary charge; relational time; Wheeler–DeWitt.

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Part I

Boundary, claim, and dependency lock

1. Why WDW-3 exists

WDW-1 constructs a connection-typed canonical presentation of the frozen Einstein–source realization. WDW-2 constructs its quotient by proper spatial redescription and preserves the complete source–realization family. Neither result calculates the bracket of two normal generators, identifies the full characteristic distribution, or proves that finite changes of Cauchy embedding are represented faithfully by the canonical constraint directions. Those are distinct burdens.

The exact WDW-3 burden. Starting from the accepted WDW-1/WDW-2 object, calculate the full classical lapse–shift algebroid; prove its regular characteristic realization; construct the corresponding history-relative embedding-change groupoid; and determine exactly when a finite change of slicing is quotient-trivial and when it carries retained source, boundary, singular, topological, or relational standing.

Three implications that are often conflated are separated throughout:

$$\begin{aligned} \text{spatial descent} &\not\Rightarrow \text{normal-normal closure}, \\ \text{normal-normal closure} &\not\Rightarrow \text{regular characteristic equality}, \\ \text{regular characteristic equality} &\not\Rightarrow \text{global finite refoliation}. \end{aligned}$$

The paper proves the first missing implication by direct functional calculation, the second by an elliptic split-rank argument, and the locally valid part of the third by a constructed history-embedding groupoid. It classifies rather than erases global obstructions.

2. Exact claim and exact nonclaims

Definition 2.1 (Passing branch). A branch b is *passing at WDW-3* when it is eligible under the accepted WDW-1 and WDW-2 releases and all WDW-3 certificates used by the asserted conclusion are populated. Algebraic closure, regular reduction, finite history reconstruction, and boundary extension are separate coordinates. Passing one never silently populates another.

Theorem 2.2 (Reader-facing WDW-3 theorem). *On the occupied compact boundaryless vacuum regular branch, the complete lapse–shift constraint family represents the hypersurface-deformation Lie algebroid off shell on a declared smooth common domain. The full constraint zero locus is split tame near an explicit no-KID point; its characteristic distribution is precisely the distribution generated by proper tangential and normal constraints; and the latter is locally integrated by a typed groupoid of admissible Cauchy-embedding changes in one reconstructed history. Proper refoliation preserves source fibre, attachment, reconstruction class, Dirac and relational profiles, and boundary certificate. KID isotropy, large transformations, charge-bearing boundaries, changed sources, changed relational data, and target-changing routes are retained.*

The massive minimally coupled Klein–Gordon branch satisfies the same exact off-shell constraint algebra. Its regular-reduction and finite-refoliation conclusions hold at, and only at, points carrying a populated ScalarReg3 certificate.

Nonclaim 2.3 (Excluded stronger readings). No theorem below asserts:

1. a fixed Lie algebra with constant structure coefficients;
2. a global Hamiltonian group action of refoliations on instantaneous phase space;

3. a smooth Hausdorff global superspace;
4. a differentiable unreduced lapse–shift action or flow on one fixed Hilbert cotangent bundle;
5. regularity at KID points or unrestricted regularity for massive scalar data;
6. triviality of asymptotic or finite-boundary Hamiltonians;
7. a preferred foliation, lapse, clock, or time coordinate; or
8. a Wheeler–DeWitt operator, regulator, anomaly theorem, physical inner product, or quantum solution of the problem of time.

Definition 2.4 (Correctness-relevant objection). An objection has standing against the frozen endpoint only if it preserves the theorem target and does at least one of the following: breaks an identified import; shows a type, domain, density, sign, or bundle failure; invalidates the occupied regular point or split rank; supplies a same-scope deformation route outside the residual classifier; or shows that source, boundary, singular, large, or relational data have been wrongly collapsed.

3. Frozen target and branch table

The core target fixes a smooth compact connected oriented three-manifold Σ without boundary, Lorentzian signature $(- + +)$, the WDW-1 positive extrinsic-curvature convention, the WDW-2 smooth tame carrier and Sobolev tower, the full smearing spaces

$$\mathcal{X} = \Gamma^\infty(T\Sigma), \quad \mathcal{N} = C^\infty(\Sigma),$$

and globally hyperbolic reconstructed histories with cooriented spacelike Cauchy embeddings. The identity-component spatial action groupoid is retained. A selected lapse is never substituted for $\mathcal{N}$.

Branch	Algebra	Regular/finite result	Status
Vacuum	Direct off-shell three-bracket closure	Occupied no-KID chart; local finite refoliation; singular KID retention	PROVED/OCCUPIED
Massive Klein–Gordon pointed lift	Direct off-shell total three-bracket closure	At each populated ScalarReg3 point and licensed reconstructed history	PROVED ALGEBRA; CONDITIONAL QUOTIENT
Internal-gauge extension	Only after Gauss, lift, domain, boundary, global-form, and extension-bracket data pass InternalCert3	Only after generalized KID rank and history certificates pass	CONDITIONAL
Unresolved internal branch	Not admitted	Not admitted	BLOCKED
Noncanonical forced source	No source-faithful total canonical normal generator	Not admitted	BLOCKED BY WDW-1

Proposition 3.1 (Scalar branch discipline). *The scalar algebra theorem is independent of scalar linearization stability. The scalar regular quotient is not.*

Proof. The algebra calculation in section 19 uses only the canonical scalar functional, its common differentiability domain, and the gravitational canonical variables. Regular reduction additionally requires closed range and a right inverse for the coupled linearized constraint map, equivalently the absence of generalized matter KIDs under the stated analytic hypotheses. The published scalar result [1] has an erratum [2]; it therefore cannot populate the latter coordinate without an independent pointwise certificate. $\square$

The compact boundaryless branch is the only occupied boundary class. For a boundary extension the complete differentiable generator is

$$G_{\text{tot}}[\xi, N] = \mathcal{D}[\xi] + H_{\perp}[N] + B[\xi, N].$$

The arrow is proper only when the allowed variation is differentiable and the resulting physical charge vanishes on the declared boundary phase space. A nonzero boundary Hamiltonian is physical evolution or a physical symmetry, not a refoliation to be removed.

4. Dependency order

Let G_1 and G_2 denote the exact accepted WDW-1 and WDW-2 gates. WDW-3 is the acyclic chain

$$G_1 \longrightarrow G_2 \longrightarrow G_{3,\text{alg}} \longrightarrow G_{3,\text{reg}} \longrightarrow G_{3,\text{emb}} \longrightarrow G_{3,\text{fin}} \longrightarrow G_3.$$

Here $G_{3,\text{alg}}$ is branchwise and off shell; $G_{3,\text{reg}}$ is pointwise; $G_{3,\text{emb}}$ requires a licensed reconstructed history; and $G_{3,\text{fin}}$ is local on regular embedding charts with all global residuals guarded. No downstream theorem repairs a missing upstream antecedent.

Proposition 4.1 (No reverse authorization). *Neither successful finite comparison nor AASC standing closure can supply an absent canonical bracket, constraint-map rank, history existence theorem, or boundary differentiability condition.*

Proof. Each downstream object is typed over the preceding one. The embedding functor is defined only on evaluated constraint data, regular characteristic leaves require the split zero locus, and the AASC comparison requires already determinate carriers and admissible maps. Removing an upstream type makes the downstream expression undefined rather than retroactively true. This is the same no-repair dependency principle imported from the kernel and capstone, applied here to the conventional mathematical carrier [3, 4]. $\square$

5. Import-fidelity ledger

The publication imports are:

Object	Role	Use in WDW-3
WDW-1	Canonical reconstruction and branch eligibility	Canonical variables, source typing, and the boundary guard
WDW-2	Proper spatial action-groupoid quotient	Spatial/internal descent, regular slices, and the equivariant lapse family
WDW-3 contract	Refoliation endpoint specification	Typed target, scope boundary, and downstream handoff

The load-bearing conventional imports are deliberately narrower than their surrounding literature:

1. Teitelboim supplies the geometric deformation calculation and embeddability interpretation, after translation to the frozen signs [5].
2. Hojman–Kuchař–Teitelboim supplies only a conditional uniqueness comparison under its own locality, variable, and time-reversibility assumptions [6].
3. Blohmann–Fernandes–Weinstein supplies the Σ -evolution groupoid and its Lie algebroid, while explicitly leaving the direct phase-space connection open [7].
4. Fischer–Marsden–Moncrief supplies the compact-CMC relation among KIDs, linearization stability, smooth regular points, and conical singular behavior [8].

5. Beig–Chruściel–Schoen supplies no-KID genericity only in its stated compact-CMC, asymptotically flat, or asymptotically hyperbolic regimes [9]. It is not used as occupancy.
6. Choquet-Bruhat–Geroch supplies uniqueness of maximal globally hyperbolic development, while Bernal–Sánchez supplies smooth Cauchy splitting [10–12]. Existence of many splittings supplies no preferred one.
7. Regge–Teitelboim and Pons–Salisbury–Shepley govern boundary differentiability and projectable descriptor calibration, respectively [13, 14].
8. Cattaneo and Blohmann–Schiavina–Weinstein supply conormal/coisotropic and Lie–Rinehart structural comparison only; the WDW-3 mixed-order chart remains an internal burden [15, 16].
9. Chruściel–Delay supplies conventional constraint mapping-property comparison. Its unequal mixed row weights are preserved explicitly; its weighted noncompact theorem is not substituted for the compact parameter estimates proved here [17].
10. Crainic–Fernandes is retained only as finite-dimensional integration comparison. It supplies no Banach or Fréchet Lie-II theorem used by WDW-3 [18].
11. Taylor supplies only the compact elliptic pseudodifferential parametrix, regularity, and Fredholm theorem. The row balancer, its principal symbol, parameter estimates, and conversion back to a right inverse of the original constraint map are proved in W3.4 [19].
12. Hamilton’s Nash–Moser theorem is invoked only after W3.6 displays the shifted tower and proves a smooth-tame family of two-sided inverses to differentiated evaluation [20].

The AASC imports have a different role. The kernel fixes the necessity of admissibility, standing, reference, and irreversible fixation for determinate carriers. The physical-clock theorem classifies clock use. The Gauge Transition Groupoid theorem supplies boundary-fixed local-presentation and no-selector grammar. The unification modules preserve source fibres and attachment ancestry. None calculates the ADM brackets, creates a KID-free point, or proves a Cauchy theorem [3, 21–23].

Import discipline. No theorem in this manuscript is imported from a corpus matrix summary. Exact excerpts, hypotheses, specializations, prohibited promotions, and downstream consumers are recorded in the bundled machine ledger.

Part II

Inherited canonical and groupoid object

6. The WDW-1 canonical reconstruction object

For each passing branch b , WDW-1 supplies a smooth canonical carrier $\mathcal{P}_b^\infty$, a weak canonical two-form Ω_b , constraint candidates, and a source-attached history family. The vacuum carrier is

$$\mathcal{P}_g^\infty = \text{Met}^\infty(\Sigma) \times \Gamma^\infty(S^2T\Sigma \otimes \text{Dens } \Sigma),$$

with coordinates (h_{ij}, π^{ij}) . The scalar pointed lift is

$$\mathcal{P}_\phi^\infty = \mathcal{P}_g^\infty \times C^\infty(\Sigma) \times \Gamma^\infty(\text{Dens } \Sigma),$$

with coordinates $(h, \pi; \phi, p_\phi)$. On either branch

$$\Omega_b((u, \alpha), (v, \beta)) = \langle \alpha, v \rangle - \langle \beta, u \rangle, \quad \iota_{X_F} \Omega_b = -dF.$$

The gravitational densities are

$$\mathcal{H}_\perp^g = \frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{2\kappa} (R - 2\Lambda), \quad (1)$$

$$\mathcal{H}_i^g = -2D_j \pi^j_i. \quad (2)$$

For the exact minimally coupled massive scalar lift,

$$\mathcal{H}_\perp^\phi = \frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \frac{\sqrt{h}}{2} m^2 \phi^2, \quad (3)$$

$$\mathcal{H}_i^\phi = p_\phi D_i \phi. \quad (4)$$

The total scalar-branch constraints are the sums of the gravitational and matter densities. The scalar is a genuine canonical lift; a source history without such a lift does not acquire a Hamiltonian generator by notation.

The sign and density lock is

$$(- + ++), \quad K_{ij} = \frac{1}{2} \mathcal{L}_n h_{ij}, \quad \pi^{ij} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij}),$$

and

$$[\nabla_\mu, \nabla_\nu] v^\rho = R^\rho_{\sigma\mu\nu} v^\sigma.$$

Consequently

$$K_{ij} = \frac{2\kappa}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right)$$

and the geometric vacuum Hamiltonian constraint is

$$R + K^2 - K_{ij} K^{ij} - 2\Lambda = 0.$$

All external formulas are translated to this convention before use.

Definition 6.1 (Structured source coordinate). The WDW-1 source coordinate at a complete history is

$$(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}),$$

where $\hat{x}$ is the admitted Module-A quotient representative, $\text{Att}_{\hat{x}}$ is the attachment witness, and $W_{\hat{x}}$ carries the reconstruction and Ward data. Equality of stress-energy tensors alone is not equality of this coordinate.

7. The WDW-2 spatial action groupoid

WDW-2 uses the smooth Fréchet Lie group

$$\mathcal{G}_{\text{sp}}^\infty = \text{Diff}_0^\infty(\Sigma)$$

on the smooth regular cotangent carrier. The full action is not placed on one fixed T^*H^s completion. Its declared left action has fundamental vector

$$\rho_{\text{left}}(\xi)q = -\mathcal{L}_\xi q,$$

while the conventional positive-Lie-drag generator is

$$\mathcal{D}_b[\xi] = \int_\Sigma \left(\pi^{ij} \mathcal{L}_\xi h_{ij} + p_\phi \mathcal{L}_\xi \phi \right).$$

The left-action moment map is therefore

$$\langle J_{\text{left}}, \xi \rangle = -\mathcal{D}_b[\xi], \quad \iota_{\rho_{\text{left}}(\xi)} \Omega_b = -d\langle J_{\text{left}}, \xi \rangle = d\mathcal{D}_b[\xi].$$

The conventional Hamiltonian vector field satisfies

$$X_{\mathcal{D}[\xi]} = -\rho_{\text{left}}(\xi).$$

This separation prevents the sign error produced by calling the same map both the left-action anchor and the positive-Lie-drag generator.

Let

$$\mathcal{C}_{\text{sib}} = J_{\text{left},b}^{-1}(0).$$

For a canonical presentation cover $\{U_a\}$, WDW-2 forms

$$\widetilde{\mathcal{C}}_{\text{sib}} = \coprod_a (U_a \cap \mathcal{C}_{\text{sib}})$$

and the phase/action groupoid

$$\mathbb{A}_{\text{prop}_b} = \mathbb{G}_{\text{pres},b} \ltimes \widetilde{\mathcal{C}}_{\text{sib}} \rightrightarrows \widetilde{\mathcal{C}}_{\text{sib}}.$$

An arrow (g, z) has

$$\begin{aligned} s(g, z) &= (a, z), & t(g, z) &= (c, g \cdot z), \\ (k, g \cdot z) \circ (g, z) &= (k \circ g, z), & (g, z)^{-1} &= (g^{-1}, g \cdot z). \end{aligned}$$

Its quotient notation

$$\mathcal{C}_{\text{sib}} // \mathbb{A}_{\text{prop}_b}$$

denotes the groupoid/Morita object characterized by the invariant-map universal property, not an assumed global Hausdorff orbit manifold.

Proposition 7.1 (WDW-2 import fidelity). *The spatial groupoid used below is exactly the accepted WDW-2 action groupoid. Its regular local quotient charts, singular objects, cover augmentation, source-history descent, and boundary and large-transformation guards are unchanged.*

Proof. The objects, arrows, action anchor, source and target, composition, inverse, and guard predicates are the definitions imported above. WDW-3 adds the normal characteristic relation only after taking their product-compatible history correspondence; it neither replaces the groupoid by one slice nor enlarges its proper-arrow predicate. The exact import is recorded in the WDW-2 publication interface of section 5. $\square$

8. The equivariant normal family

For $N \in \mathcal{N}$, define

$$H_{\perp,b}[N] = \int_{\Sigma} N \mathcal{H}_{\perp,b}.$$

WDW-2 proves the equivariance relation

$$\{\mathcal{D}_b[\xi], H_{\perp,b}[N]\} = H_{\perp,b}[\mathcal{L}_{\xi}N]. \quad (5)$$

WDW-3 rederives it in section 17 as a sign and density regression check.

A fixed smeared functional $H_{\perp}[N]$ is not basic on the spatial quotient when N is held numerically fixed while the fields are pulled back. The invariant object is the equivariant family

$$N \longmapsto H_{\perp}[N], \quad (g, N) \longmapsto (g \cdot z, g_*N).$$

Thus WDW-3 receives all lapse directions, not one chosen evolution.

Lemma 8.1 (No fixed-lapse substitution). *Knowledge of $H_{\perp}[1]$, or of $H_{\perp}[N_0]$ for one positive N_0 , does not determine the normal-normal bracket as a local constraint identity.*

Proof. The target bracket contains

$$\beta_h(N, M)^i = h^{ij}(ND_jM - MD_jN),$$

which vanishes when both smearings are constant. A fixed-lapse calculation therefore deletes precisely the derivative data that tests the structure function and embeddability. $\square$

9. Source, reconstruction, observable, and boundary coordinates

The complete inherited object is

$$\mathfrak{H}_{W2}(b) = (\mathcal{C}_{\text{si}b} // \mathbb{A}_{\text{prop}b}, \mathfrak{A}_b^{\text{reg}}, \overline{\mathcal{O}}_{\text{prop},b}, \overline{\text{Src}\mathfrak{M}}, \\ \overline{\text{Recon}}_b, H_{\perp,b}[\mathcal{N}], \text{Emb}_b, \text{BdyCert}_b, \text{Open}_{G_3,b}),$$

with the regular-chart family

$$\mathfrak{A}_b^{\text{reg}} = \left\{ T_{\text{reg}}^* S_a^\infty, \omega_{\text{red},a}^\dagger, T^* S_a^r, \omega_{\text{red},a}^r \right\}_{a,r \geq r_0}.$$

The smooth reduced form is weakly symplectic. Each completed slice $T^* S_a^r$ carries a strong Hilbert symplectic form. No compatibility theorem for completed overlap maps is imported or asserted.

Lemma 9.1 (No tensor-only reconstruction). *If two canonical histories have the same bare stress-energy tensor but different $(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}})$, WDW-3 does not identify them.*

Proof. The source triple is a coordinate of the object space and all groupoid arrows are required to preserve it. Equality of one downstream tensor does not supply an arrow between different fibres. This is both the WDW-1 source-attachment rule and the AASC standing-bearing tensor rule. $\square$

The observable coordinate is also stratified:

1. Dirac observables are constant on the full characteristic relation;
2. relational observables are correlations on the complete reconstructed history;
3. records carry their readout and retention data;
4. boundary observables may generate nonzero Hamiltonians; and
5. presentation variables include lapse, shift, cover label, and a representative slice only while the standing-bearing coordinates are fixed.

10. Analytic carrier and common domain

All unreduced brackets are computed on the smooth inverse-limit carrier

$$\mathcal{P}_b^\infty = \varprojlim_{s \geq s_0} \mathcal{P}_b^s.$$

The model Sobolev levels are

$$h, \phi \in H^s, \quad \pi, p_\phi \in H^{s-1}, \quad s > \frac{7}{2},$$

with the momentum variables interpreted in their density bundles. The constraint map has the level behavior

$$\Phi_b : \mathcal{P}_b^s \longrightarrow H^{s-2}(\text{Dens } \Sigma) \oplus H^{s-2}(T^* \Sigma \otimes \text{Dens } \Sigma).$$

The curvature term loses two derivatives; the divergence of momentum loses one; Lie transport and composition are handled one level higher when differentiated. The smooth tower accommodates these shifts. No smooth action or normal flow is claimed on the full fixed-index carrier.

Definition 10.1 (WDW-3 common functional algebra). $\mathcal{A}_{3b}$ is the algebra of smooth regular local functionals on $\mathcal{P}_b^\infty$ such that:

1. first variational derivatives are smooth sections of the declared dual density bundles;
2. all integrations by parts are licensed on the compact core;

3. $\mathcal{D}[\xi]$, $H_\perp[N]$, and every first mutual bracket lie in the algebra;
4. their Hamiltonian vector fields are tangent to the smooth carrier;
5. gravitational and matter terms share the same domain; and
6. the functional obtained is independent of whether one expands before or after the licensed integrations by parts.

Proposition 10.2 (Common-domain closure). *For smooth $h, \pi, \phi, p_\phi, \xi, N, M$ on compact boundaryless Σ , the constraint functionals in equations (1) to (4), their first variational derivatives, and the three brackets calculated in part IV belong to $\mathcal{A}_{3b}$.*

Proof. Every density is polynomial in $\pi, p_\phi, h^{\pm 1}$, and at most two derivatives of h or one derivative of ϕ . Smooth positive-definite metrics make inversion and $\sqrt{h}$ smooth tame maps. Compactness removes boundary remainders. The explicit calculations below show that second derivatives of smearings recombine, after two integrations by parts, into a first-order momentum generator; no third derivative of a canonical field remains. The scalar cross terms are algebraic in the smearings and cancel antisymmetrically. These facts give closure and route independence. $\square$

Nonclaim 10.3 (Flow boundary). The existence of Hamiltonian vector fields as smooth derivations on $\mathcal{P}_b^\infty$ is not promoted to a global Fréchet ODE theorem. Local spacetime evolution is obtained from a separately gauge-fixed hyperbolic Cauchy problem. Sobolev levels supply estimates, elliptic splittings, and local completed slices, not a universal normal Hamiltonian flow on $T^*S_a^s$.

Theorem W3.1 (Exact inherited object, branch boundary, and common domain). The WDW-3 input is the full tuple $\mathfrak{H}_{W2}(b)$ on exactly the branch table of definition 2.1. Its smooth carrier, mixed Sobolev tower, proper spatial action groupoid, source and reconstruction coordinates, equivariant lapse family, singular objects, and boundary guard are jointly well typed on $\mathcal{A}_{3b}$. No fixed lapse, global Hilbert action carrier, tensor-only source reconstruction, or local-slice substitution is admitted.

Proof. Sections 6–10 give each component, its type, and its governing accepted import. Propositions 7.1 and 10.2 and lemmas 8.1 and 9.1 prove the four non-substitution clauses. The publication interfaces and branch statuses are fixed in section 5. $\square$

Part III

Geometric hypersurface-deformation algebroid

11. Admissible embedded histories

Definition 11.1 (Licensed reconstructed history). A licensed history object on branch b is

$$\mathbf{H} = (M_\gamma, g_\gamma, i, \mathcal{F}; \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, \text{BdyCert}_b),$$

where:

1. (M_γ, g_γ) is a time-oriented globally hyperbolic solution in the WDW-1 reconstruction class;
2. $i : \Sigma \hookrightarrow M_\gamma$ is a smooth cooriented spacelike Cauchy embedding;
3. $\mathcal{F}$ is a local foliation presentation containing $i(\Sigma)$;
4. the source triple is the inherited branch coordinate; and
5. the boundary certificate is the occupied compact-core certificate or a separately populated extension.

For vacuum data the maximal globally hyperbolic development exists and is unique up to isometry in the relevant class [10]. Smooth globally hyperbolic spacetimes admit smooth spacelike Cauchy hypersurfaces and smooth Cauchy splittings [11, 12]. These theorems license history comparison. They do not select a preferred embedding or prove that arbitrary canonical data satisfy the constraints.

Lemma 11.2 (Existence/comparison separation). *An arrow comparing two embeddings in one licensed history presupposes that history; it does not prove existence of a development for a new constraint datum.*

Proof. The arrow type contains γ as a common coordinate. Removing γ leaves no codomain for either embedding and no evaluation map. Existence is therefore an antecedent, not a consequence, of the groupoid comparison. $\square$

12. The lapse–shift deformation bundle

At (γ, i) , the deformation descriptor is

$$e = (\xi, N) \in \mathcal{E}_{\text{def}(\gamma, i)} := \Gamma^\infty(T\Sigma) \oplus C^\infty(\Sigma).$$

With future unit normal n fixed by the coorientation, its physical forward velocity on embeddings is

$$v_{\text{emb}}(\xi, N)_i = i_*\xi + Nn \in \Gamma^\infty(i^*TM_\gamma). \quad (6)$$

The tangential component changes the parametrization of the embedded leaf. The normal component changes the embedded leaf in the same spacetime. Both components are presentation directions only after the complete properness and standing guards pass.

The physical velocity and the Lie-algebroid anchor carry different signs. In the right/fundamental convention inherited from the history groupoid, set

$$\rho_{\text{src}}(\xi, N)_i = -v_{\text{emb}}(\xi, N)_i. \quad (7)$$

The physical embedding fields form an anti-representation of the displayed deformation bracket, whereas ρ_{src} is an ordinary Lie-algebroid anchor. This sign is kept visible through the canonical comparison in section 21.

The bundle is not a product Lie algebra over instantaneous phase space. Its bracket depends on the induced metric

$$h = i^*g_\gamma.$$

The dependence is geometrically necessary: two successive normal displacements fail to commute by a tangential displacement whose coefficient uses h^{-1} .

13. Geometric deformation bracket

Choose Gaussian normal coordinates near $i(\Sigma)$ and extend (ξ, N) by the Gaussian rule that preserves the normal decomposition at the reference leaf. A direct commutator calculation gives

$$[(\xi, N), (\eta, M)]_h = ([\xi, \eta] + N \text{grad}_h M - M \text{grad}_h N, \mathcal{L}_\xi M - \mathcal{L}_\eta N). \quad (8)$$

The three pure cases are

$$[(\xi, 0), (\eta, 0)]_h = ([\xi, \eta], 0), \quad (9)$$

$$[(\xi, 0), (0, M)]_h = (0, \mathcal{L}_\xi M), \quad (10)$$

$$[(0, N), (0, M)]_h = (\beta_h(N, M), 0), \quad (11)$$

where

$$\beta_h(N, M)^i = h^{ij}(ND_j M - MD_j N). \quad (12)$$

This is the Lorentzian-sign bracket in the frozen normal convention. Reversing the normal or using $K = -\frac{1}{2}\mathcal{L}_n h$ changes the comparison signs and must be translated globally, not piecemeal.

Theorem W3.2 (Geometric hypersurface-deformation algebroid). Over the space of cooriented spacelike embedded-history objects, the bundle $\mathcal{E}_{\text{def}}$ with anchor ρ_{src} and bracket equation (8) is a Lie algebroid.

Proof. The bracket is obtained from commutators of the corresponding Gaussian extensions. In the right/fundamental convention,

$$[v_{\text{emb}}(e_1), v_{\text{emb}}(e_2)] = -v_{\text{emb}}([e_1, e_2]_h),$$

so $\rho_{\text{src}} = -v_{\text{emb}}$ preserves brackets. Commutators supply antisymmetry and Jacobi, including the variation of the induced metric. The Gaussian extension is not closed under bracket as a fixed subspace of vector fields, but its value at the reference leaf and the anchor correction give the section bracket. For $f \in C^\infty$ of the base,

$$[e_1, f e_2]_h = f[e_1, e_2]_h + (\rho_{\text{src}}(e_1)f)e_2.$$

Metric dependence in β_h is differentiated by the anchor in the Jacobi identity. This is precisely the Lie-algebroid construction of Blohmann et al. [7, Theorem 2.12], restricted here to licensed reconstructed histories and written in the right/fundamental presentation for which equation (6) remains the physical forward velocity. $\square$

Remark 13.1 (Why freezing h fails). If one holds h fixed and treats equation (8) as a bracket on $\mathcal{X} \oplus \mathcal{N}$, its Jacobiator is generally nonzero. The missing terms are exactly the variations of h^{ij} under the anchors. The structure is a Lie algebroid, not an infinite-dimensional Lie algebra with numerical structure constants.

14. Signature and embeddability

The sign in the normal–normal component is not decorative. Let $e_N = Nn$ and $e_M = Mn$. Their Gaussian extensions obey

$$[e_N, e_M]|_{i(\Sigma)} = i_*(N \text{grad}_h M - M \text{grad}_h N).$$

For a timelike unit normal with $g(n, n) = -1$, this is the sign in equation (8). A spacelike-normal theory has the opposite signature factor. Teitelboim’s embeddability calculation identifies this commutator as the geometric content of the canonical normal–normal relation [5].

Definition 14.1 (Embeddable canonical representation). A canonical descriptor representation is embeddable on a licensed history class when:

1. its spatial, mixed, and normal brackets agree with equation (8) after one explicit anchor convention is fixed;
2. its Hamiltonian vector fields equal the differential of evaluation on that class; and
3. its finite local characteristic leaves agree with local embedding-change orbits.

The first condition is proved in part IV; the second in section 34; and the third in section 35. Merely naming the canonical formula the “Dirac algebra” does not prove any of the three.

15. Comparison with the Σ -evolution groupoid

Let $\mathbb{E}_\Sigma^{\text{BFW}}$ denote the groupoid whose objects are Σ -universes—*isometry classes of cooriented spacelike embeddings*—and whose arrows are pairs of embeddings into a common Lorentzian target. Its orbits consist of embeddings in a fixed spacetime, its isotropy records spacetime isometries, and its Lie algebroid has fibre $\mathcal{X} \oplus \mathcal{N}$ with bracket equation (8) [7].

That result also states its own limit: the algebroid lies over a space larger than ADM phase space, and the paper does not construct a direct connection from the groupoid to the Hamiltonian constraint

functions. The difficulty is not erased by observing that the two brackets look identical. A map may preserve a displayed formula while failing to be defined on the required objects.

Definition 15.1 (WDW-3 restricted comparison object). $\mathbb{E}_b$ is the full subgroupoid of the augmented Σ -evolution construction whose objects:

1. lie in a WDW-1 reconstructed history of branch b ;
2. evaluate to $\mathcal{C}_{\text{tot}b}$;
3. carry the inherited source, attachment, witness, boundary, and relational records;
4. obey the WDW-2 proper spatial guard; and
5. lie in the declared regular chart when a manifold or local-integration conclusion is asserted.

WDW-3 closes the missing interface by constructing directly

$$\text{Eval}^{\text{path}} : \Pi_1^{\text{guard}}(\mathcal{O}_{\text{emb}}) \longrightarrow \mathbb{C}_{b,U}^{\text{char}}$$

and proving

$$d\text{Eval}(v_{\text{emb}}(\xi, N)) = X_{\mathcal{D}[\xi] + H_{\perp}[N]},$$

equivalently

$$d\text{Eval} \circ \rho_{\text{src}} = \rho_{\text{con}} \circ \Psi$$

for the on-shell conormal morphism Ψ constructed below. The external groupoid theorem supplies the geometry of the source object. The canonical functional calculation, coefficient-sensitive morphism, shifted Sobolev evaluation theorem, and path/homotopy functor laws are new interface obligations discharged within this manuscript; they are not attributed to the external source or to a finite-dimensional Lie-II theorem.

Part IV

Canonical bracket closure

16. Direct spatial–spatial regression check

The Poisson convention inherited from WDW-1 is

$$\{F, G\} = dF(X_G) = \left\langle \frac{\delta F}{\delta q}, \frac{\delta G}{\delta p} \right\rangle - \left\langle \frac{\delta F}{\delta p}, \frac{\delta G}{\delta q} \right\rangle, \quad \iota_{X_F} \Omega = -dF.$$

For

$$\mathcal{D}_b[\xi] = \int_{\Sigma} \left(\pi^{ij} \mathcal{L}_{\xi} h_{ij} + p_{\phi} \mathcal{L}_{\xi} \phi \right),$$

compactness gives

$$\delta \mathcal{D}_b[\xi] = \int_{\Sigma} \left(\delta \pi^{ij} \mathcal{L}_{\xi} h_{ij} - (\mathcal{L}_{\xi} \pi^{ij}) \delta h_{ij} + \delta p_{\phi} \mathcal{L}_{\xi} \phi - (\mathcal{L}_{\xi} p_{\phi}) \delta \phi \right).$$

Thus

$$X_{\mathcal{D}[\xi]}(h, \pi; \phi, p_{\phi}) = (\mathcal{L}_{\xi} h, \mathcal{L}_{\xi} \pi; \mathcal{L}_{\xi} \phi, \mathcal{L}_{\xi} p_{\phi}).$$

Direct substitution into the functional bracket gives

$$\{\mathcal{D}_b[\xi], \mathcal{D}_b[\eta]\} = \mathcal{D}_b[[\xi, \eta]]. \quad (13)$$

There is no surface cocycle on the compact core.

Remark 16.1 (Function and vector-field conventions). The function assignment $\xi \mapsto \mathcal{D}[\xi]$ obeys equation (13). With $\iota_{X_F}\Omega = -dF$,

$$[X_F, X_G] = -X_{\{F, G\}}.$$

Hence the corresponding fundamental vector fields form the expected anti-representation. This is the same left/right convention issue already isolated by the WDW-2 moment-map repair. All later statements distinguish Poisson-bracket preservation from the ordinary commutator sign of fundamental vector fields.

17. Direct spatial–normal regression check

The total normal density is a scalar density of weight one. Under the positive-Lie-drag Hamiltonian field of $\mathcal{D}[\xi]$,

$$\delta_\xi \mathcal{H}_\perp = \mathcal{L}_\xi \mathcal{H}_\perp = D_i(\xi^i \mathcal{H}_\perp).$$

Keeping the smearing as an argument of the equivariant family and integrating by parts,

$$\begin{aligned} \{\mathcal{D}[\xi], H_\perp[N]\} &= \int_\Sigma N \mathcal{L}_\xi \mathcal{H}_\perp \\ &= - \int_\Sigma (\mathcal{L}_\xi N) \mathcal{H}_\perp \quad \text{with the opposite order of the functional variation.} \end{aligned}$$

In the frozen ordered convention this is

$$\boxed{\{\mathcal{D}[\xi], H_\perp[N]\} = H_\perp[\mathcal{L}_\xi N].} \quad (14)$$

Equivalently $\{H_\perp[N], \mathcal{D}[\xi]\} = -H_\perp[\mathcal{L}_\xi N]$. Direct differentiation of equations (1) and (3) gives the same formula: the curvature is a scalar, π and p_ϕ have weight one, and all connection variations combine into the Lie derivative of the total density. This reproduces the accepted WDW-2 sign rather than importing it by familiarity.

18. Vacuum normal–normal bracket

Write

$$\begin{aligned} H_g[N] &= T[N] + V[N], \\ T[N] &= \int_\Sigma \frac{2\kappa N}{\sqrt{h}} \left(\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2 \right), \quad V[N] = - \int_\Sigma \frac{N\sqrt{h}}{2\kappa} (R - 2\Lambda). \end{aligned}$$

The momentum derivative is

$$\frac{\delta H_g[N]}{\delta \pi^{ij}} = \frac{4\kappa N}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) := N A_{ij}. \quad (15)$$

At fixed contravariant density π^{ij} , the metric derivative has the form

$$\frac{\delta H_g[N]}{\delta h_{ij}} = N Q^{ij} - \frac{\sqrt{h}}{2\kappa} \left(D^i D^j N - h^{ij} D^2 N \right), \quad (16)$$

where Q^{ij} is a local tensor density containing the kinetic metric derivative, Einstein tensor, and cosmological term, but no derivative of N . For completeness,

$$\begin{aligned} Q^{ij} &= \frac{2\kappa}{\sqrt{h}} \left(2\pi^i{}_k \pi^{jk} - \pi \pi^{ij} - \frac{1}{2} h^{ij} \left(\pi_{kl} \pi^{kl} - \frac{1}{2} \pi^2 \right) \right) \\ &\quad + \frac{\sqrt{h}}{2\kappa} \left(R^{ij} - \frac{1}{2} h^{ij} R + \Lambda h^{ij} \right). \end{aligned}$$

The bracket splits into four pieces. Because $T[N]$ is ultralocal and linear in N ,

$$\{T[N], T[M]\} = 0.$$

Because V contains no momentum,

$$\{V[N], V[M]\} = 0.$$

The terms involving NQ^{ij} and MA_{ij} are proportional to NM and cancel under antisymmetrization. The only surviving terms pair equation (15) with the smearing-derivative part of equation (16):

$$\begin{aligned} \{H_g[N], H_g[M]\} &= \int_{\Sigma} \left[-\frac{\sqrt{h}}{2\kappa} (D^i D^j N - h^{ij} D^2 N) \right] \left[\frac{4\kappa M}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) \right] \\ &\quad - (N \leftrightarrow M). \end{aligned}$$

The trace pieces cancel inside the contraction:

$$(D^i D^j N - h^{ij} D^2 N) \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) = \pi^{ij} D_i D_j N.$$

Therefore

$$\{H_g[N], H_g[M]\} = 2 \int_{\Sigma} \pi^{ij} (N D_i D_j M - M D_i D_j N). \quad (17)$$

Now

$$D_i \beta_j(N, M) = (D_i N)(D_j M) + N D_i D_j M - (D_i M)(D_j N) - M D_i D_j N.$$

Symmetry of π^{ij} cancels the two gradient products, so

$$\int_{\Sigma} \pi^{ij} \mathcal{L}_{\beta} h_{ij} = 2 \int_{\Sigma} \pi^{ij} D_i \beta_j$$

equals the right side of equation (17). Hence

$$\boxed{\{H_g[N], H_g[M]\} = \mathcal{D}_g[\beta_h(N, M)]}. \quad (18)$$

No constraint was imposed at any step. This is an off-shell identity on $\mathcal{A}_{3g}$.

Proposition 18.1 (Vacuum cancellation audit). *Every term omitted from equation (17) is either kinetic-kinetic, potential-potential, or symmetric in NM ; none is set to zero by the constraints.*

Proof. The first two classes vanish for the structural reasons stated above. All remaining ultralocal metric derivatives have the form NQ^{ij} , while $\delta H[M]/\delta \pi^{ij} = MA_{ij}$. Their product is $NMQ^{ij}A_{ij}$ and is identical after $N \leftrightarrow M$. The curvature variation is the unique source of derivatives of a smearing and is retained in full. $\square$

19. Klein–Gordon matter bracket

For the scalar normal functional,

$$H_{\phi}[N] = \int_{\Sigma} N \left(\frac{p_{\phi}^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} D_i \phi D^i \phi + \frac{\sqrt{h}}{2} m^2 \phi^2 \right),$$

the canonical derivatives are

$$\frac{\delta H_{\phi}[N]}{\delta p_{\phi}} = \frac{N p_{\phi}}{\sqrt{h}}, \quad (19)$$

$$\frac{\delta H_{\phi}[N]}{\delta \phi} = -D_i (N \sqrt{h} D^i \phi) + N \sqrt{h} m^2 \phi. \quad (20)$$

Substitution gives

$$\begin{aligned} \{H_\phi[N], H_\phi[M]\} &= \int_\Sigma \left[-D_i(N\sqrt{h}D^i\phi) + N\sqrt{h}m^2\phi \right] \frac{Mp_\phi}{\sqrt{h}} \\ &\quad - \frac{Np_\phi}{\sqrt{h}} \left[-D_i(M\sqrt{h}D^i\phi) + M\sqrt{h}m^2\phi \right]. \end{aligned}$$

The mass terms cancel. Expanding the two divergences, the common second-derivative term also cancels, leaving

$$\int_\Sigma p_\phi D^i\phi (ND_iM - MD_iN).$$

Therefore

$$\boxed{\{H_\phi[N], H_\phi[M]\} = \mathcal{D}_\phi[\beta_h(N, M)]}. \quad (21)$$

The gravitational–matter cross bracket does not vanish term by term, because $H_\phi[M]$ depends on h and $H_g[N]$ depends on π . However

$$\frac{\delta H_\phi[M]}{\delta h_{ij}} = MU^{ij}, \quad \frac{\delta H_g[N]}{\delta \pi^{ij}} = NA_{ij},$$

with U^{ij} containing no derivative of M : a scalar covariant derivative is simply $D_i\phi = \partial_i\phi$. Hence the total cross contribution is

$$\int_\Sigma NM U^{ij} A_{ij} - \int_\Sigma MN U^{ij} A_{ij} = 0.$$

Combining equations (18) and (21) gives the unconditional off-shell result

$$\boxed{\{H_{g+\phi}[N], H_{g+\phi}[M]\} = \mathcal{D}_{g+\phi}[\beta_h(N, M)]}. \quad (22)$$

This calculation neither cites nor requires scalar linearization stability.

20. Certificate-gated internal extension

Certificate 20.1 (InternalCert3). An internal branch is admitted to the WDW-3 algebra only if one object records:

1. the complete internal Gauss generator and its constraint class;
2. the actual spatial lift and global gauge-group form;
3. gauge invariance of the complete normal density;
4. every Gauss-valued, field-dependent, or nonsplit extension term in the lapse–shift bracket;
5. the common functional domain and derivative-loss levels;
6. differentiable boundary improvement and charge;
7. absence or exact form of a classical central term;
8. source, attachment, representation, and sector coordinates; and
9. closure of the complete Dirac ledger, including all second-class constraints.

The conditional conclusion is not “set the unresolved terms to zero.” It is: if the populated certificate gives a bracket $[,]_{\text{ext}}$ and an improved generator C_{int} , then the total assignment is tested against that bracket. No nontrivial internal branch is occupied here.

21. Off-shell algebra and the on-shell constraint algebroid

Define the total constraint functional

$$C_b(\xi, N) = \mathcal{D}_b[\xi] + H_{\perp, b}[N].$$

Equations (13), (14), (18) and (22) give

$$\{C_b(\xi, N), C_b(\eta, M)\} = C_b([(\xi, N), (\eta, M)]_h) \quad (23)$$

for the vacuum branch and the scalar branch, with the exact geometric bracket equation (8).

21.1 W3.3a: exact off-shell function algebra

Proposition 21.1 (Exact off-shell constraint-function algebra). *On $\mathcal{A}_{3b}$, for fixed smooth descriptor arguments, the function assignment*

$$C_b : \mathcal{E}_{\text{def}} \longrightarrow C^\infty(\mathcal{P}_b^\infty)$$

obeys equation (23) without imposing any constraint.

Proof. The three independent descriptor pairs are exactly the direct calculations above. Bilinearity gives equation (23). The descriptor arguments in this assertion are fixed sections; the metric dependence of their bracket is evaluated at the phase-space point and is differentiated in the Jacobi audit below. $\square$

Remark 21.2 (Why this does not define an off-shell anchor). For $f \in C^\infty(\mathcal{P}_b^\infty)$,

$$C_b(fe) = fC_b(e), \quad X_{C_b(fe)} = fX_{C_b(e)} + C_b(e)X_f.$$

The second term is generally nonzero away from $\mathcal{C}_{\text{tot}b}$. Thus $e \mapsto X_{C_b(e)}$ is not a pointwise $C^\infty(\mathcal{P}_b^\infty)$ -linear Lie-algebroid anchor. The off-shell identity equation (23) is retained exactly, but it is not promoted to that false stronger statement.

21.2 W3.3b: constraint ideal and conormal Lie algebroid

Let $\mathcal{A}_b$ be the declared Poisson algebra of admissible smooth local functionals, and let $\mathcal{I}_b \subset \mathcal{A}_b$ be the local ideal of functionals vanishing on $\mathcal{C}_{\text{tot}b}$, generated on a regular chart by the smeared components of Φ_b . First-class closure gives

$$\{\mathcal{I}_b, \mathcal{I}_b\} \subset \mathcal{I}_b.$$

Define the constraint Lie–Rinehart module

$$\mathfrak{A}_{\text{con}, b} := \mathcal{I}_b / \mathcal{I}_b^2 \quad \text{over} \quad \mathcal{A}_b / \mathcal{I}_b \quad (24)$$

with

$$[[F], [G]]_{\text{con}} = -[\{F, G\}], \quad \rho_{\text{con}}([F]) = X_F|_{\mathcal{C}_{\text{tot}b}}. \quad (25)$$

The minus sign is forced by $[X_F, X_G] = -X_{\{F, G\}}$.

Lemma 21.3 (Well-defined conormal structure). *The formulas in equation (25) are independent of representatives, obey the Lie–Rinehart Leibniz identity, and define an anchor into $T\mathcal{C}_{\text{tot}b}$.*

Proof. Since $\{\mathcal{I}_b, \mathcal{I}_b\} \subset \mathcal{I}_b$, a Hamiltonian vector field generated by $F \in \mathcal{I}_b$ is tangent to $\mathcal{C}_{\text{tot}b}$. Moreover,

$$\{\mathcal{I}_b^2, \mathcal{I}_b\} \subset \mathcal{I}_b^2,$$

so the bracket is independent of the representative modulo $\mathcal{I}_b^2$. If a is changed by an element of $\mathcal{I}_b$, then $X_F a = \{a, F\}$ changes by an element of $\mathcal{I}_b$; hence the anchor acts well-definedly on $\mathcal{A}_b / \mathcal{I}_b$. Finally,

$$-\{F, aG\} = (X_F a)G - a\{F, G\},$$

which is precisely

$$[[F], a[G]]_{\text{con}} = a[[F], [G]]_{\text{con}} + \rho_{\text{con}}([F])(a)[G].$$

Jacobi follows from Poisson Jacobi, and bracket preservation by the anchor follows from $[X_F, X_G] = -X_{\{F, G\}}$. $\square$

At a full regular point, the standard conormal map is

$$\vartheta : \mathcal{J}_b / \mathcal{J}_b^2 \longrightarrow N^* \mathcal{C}_{\text{tot}b}, \quad \vartheta([F]) = dF|_{\mathcal{C}_{\text{tot}b}}.$$

Split surjectivity of $D\Phi_b$ makes ϑ an isomorphism on the declared local functional class. Equivalently, every conormal covector is $D\Phi_b(z)^*e = dC_b(e)|_z$; absence of KIDs makes the representing descriptor unique. This is the usual conormal Lie algebroid of a coisotropic submanifold, specialized to the mixed-order regular constraint locus [15, 16].

21.3 W3.3c: sign, orientation, and fixed-frame comparison

Object	Formula	Role
Physical forward embedding velocity	$v_{\text{emb}}(e) = i_*\xi + Nn$	Velocity of i_ϵ as ϵ increases
Geometric deformation bracket	$[e_1, e_2]_h$ from equation (8)	$[v_{\text{emb}}(e_1), v_{\text{emb}}(e_2)] = -v_{\text{emb}}([e_1, e_2]_h)$
Source algebroid anchor	$\rho_{\text{src}}(e) = -v_{\text{emb}}(e)$	Right/fundamental groupoid convention
Path/algebroid orientation	$\dot{i} = v_{\text{emb}}(e) = \rho_{\text{src}}(-e)$	The physical forward path carries algebroid label $-e$; no hidden opposite groupoid map
Hamiltonian convention	$\iota_{X_F}\Omega = -dF$	$[X_F, X_G] = -X_{\{F, G\}}$
Canonical conormal anchor	$\rho_{\text{con}}([F]) = X_F _{\mathcal{C}_{\text{tot}}}$	Ordinary Lie-algebroid anchor
Fixed-frame comparison	$\Psi_0(e) = -[C_b(e)]$	Preserves the bracket on fixed descriptor frames
Physical forward evaluation	$d\text{Eval}(v_{\text{emb}}(e)) = X_{C_b(e)}$	Proved independently in W3.5a

Indeed,

$$[\Psi_0(e_1), \Psi_0(e_2)]_{\text{con}} = -\{[C_b(e_1), C_b(e_2)]\} = \Psi_0([e_1, e_2]_h). \quad (26)$$

This is a frame-level bracket identity. It is not yet a Lie-algebroid morphism: base-dependent coefficients require the anchor square and its Leibniz terms. Those statements are proved, without backward reference, in W3.5b after differentiated evaluation has been established.

Theorem W3.3 (Off-shell algebra, on-shell conormal representation, and frame map). On the vacuum and massive Klein–Gordon branches:

1. the fixed-descriptor constraint functions obey the exact off-shell algebra equation (23);
2. on $\mathcal{C}_{\text{tot}b}$, $\mathfrak{A}_{\text{con},b}$ is the constraint Lie–Rinehart algebra, and on every full regular chart it is the conormal Lie algebroid $N^*\mathcal{C}_{\text{tot}b}$; and
3. $\Psi_0(e) = -[C_b(e)]$ obeys the fixed-frame bracket identity equation (26). No anchor or coefficient-sensitive morphism claim is made in W3.3.

No off-shell pointwise Hamiltonian-vector-field anchor is asserted.

Proof. Item 1 is proposition 21.1. Item 2 is lemma 21.3 and the regular conormal identification. Item 3 is equation (26). The calculation

$$X_{C_b(fe)} = fX_{C_b(e)} + C_b(e)X_f$$

becomes $C^\infty(\mathcal{C}_{\text{tot}b})$ -linear after pullback because the second term vanishes there, exactly as required by the conormal construction; its comparison with the source Leibniz rule is deliberately deferred to W3.5b, which consumes W3.3 rather than completing it retroactively. $\square$

22. Jacobi and structure-function audit

Poisson Jacobi on $\mathcal{A}_{3b}$ gives

$$\{C(e_1), \{C(e_2), C(e_3)\}\} + \text{cyclic} = 0.$$

Using equation (23), the middle constraint has a descriptor whose tangential component contains h^{-1} . The outer bracket therefore differentiates both the constraint functional and that descriptor. Schematically,

$$\{C(e_1), C([e_2, e_3]_h)\} = C([e_1, [e_2, e_3]_h]_h) + C\left((\rho(e_1)h) \cdot \frac{\partial[e_2, e_3]_h}{\partial h}\right).$$

The second term is the anchor contribution. It is not optional.

Proposition 22.1 (Two-route Jacobi agreement). *The Poisson-Jacobi calculation and the geometric Lie-algebroid Jacobi calculation agree on every passing algebra branch.*

Proof. The geometric bracket is obtained from commutators of Gaussian extensions and therefore satisfies algebroid Jacobi. The canonical map preserves the bracket by equation (23). Expanding either route yields the same variations of h^{ij} in β_h :

$$\delta h^{ij} = -h^{ik}h^{j\ell}\delta h_{k\ell}.$$

These terms cancel the apparent Jacobiator obtained by freezing h . The remaining terms are Lie-derivative Jacobi and the cyclic identity for scalar smearings. Appendix D records the component organization. $\square$

Part V

Full constraint regularity and characteristic equality

23. The mixed-order total constraint map

Write $r = s - 2$ and take $r > \frac{3}{2}$. In (h, K) variables the Douglis–Nirenberg scale is

$$E_r = H^{r+2}(S^2T^*\Sigma) \oplus H^{r+1}(S^2T^*\Sigma), \quad (27)$$

$$F_r = H^r(\text{Dens } \Sigma) \oplus H^r(T^*\Sigma \otimes \text{Dens } \Sigma), \quad (28)$$

$$\hat{Z}_r = H^{r+2}(\Sigma) \oplus H^{r+2}(T\Sigma). \quad (29)$$

The components of E_r are the metric and extrinsic-curvature variations; those of F_r are the Hamiltonian and momentum constraints with the inherited equal-row grading. The space $\hat{Z}_r$ is the balanced lapse-shift multiplier scale used by the fixed pseudodifferential normal problem below. It is not obtained by changing the target grading. The canonical Legendre diffeomorphism transports E_r to the corresponding (h, π) tangent scale.

For comparison with the inherited notation, let

$$\mathcal{Y}^s = H^{s-2}(\text{Dens } \Sigma) \oplus H^{s-2}(T^*\Sigma \otimes \text{Dens } \Sigma).$$

At Sobolev level $s > \frac{7}{2}$, define

$$\Phi_b^s : \mathcal{P}_b^s \longrightarrow \mathcal{Y}^s, \quad \Phi_b^s(z) = (\mathcal{H}_{\perp b}(z), \mathcal{H}_{ib}(z)).$$

Its smooth inverse-limit restriction is Φ_b^∞ . The total constraint locus is

$$\mathcal{C}_{\text{tot}b} = (\Phi_b^\infty)^{-1}(0).$$

Since its second coordinate is the WDW-2 spatial moment-map density,

$$\mathcal{C}_{\text{tot}b} \subset \mathcal{C}_{\text{sib}}.$$

The inclusion is strict away from special branches: spatial constraint satisfaction alone does not imply the Hamiltonian constraint.

Definition 23.1 (Full regular point). A point $z \in \mathcal{C}_{\text{tot}b}$ is full regular when:

1. $D\Phi_b(z)$ has closed range at every required Sobolev level;
2. $\ker D\Phi_b(z)^* = 0$;
3. $D\Phi_b(z)$ is surjective onto $\mathcal{Y}^s$;
4. a compatible tame family of right inverses exists; and
5. the WDW-2 spatial regularity and history-reconstruction certificates also pass.

Constraint row	Metric variation	K -variation	Bounded target
Hamiltonian	order 2	order 0	$H^r(\text{Dens } \Sigma)$
Momentum	order 1	order 1	$H^r(T^*\Sigma \otimes \text{Dens } \Sigma)$

24. Linearized operator, three adjoints, and normal operator

It is convenient to display the vacuum operator in (h, K) variables. Set

$$\mathcal{H}(h, K) = R(h) + K^2 - K_{ij}K^{ij} - 2\Lambda, \quad (30)$$

$$\mathcal{M}_i(h, K) = D_j K^j_i - D_i K. \quad (31)$$

The Legendre map is an isomorphism on the nondegenerate metric carrier, so regularity in (h, K) is equivalent to regularity in (h, π) .

For a variation (γ_{ij}, k_{ij}) ,

$$\begin{aligned} D\mathcal{H}(\gamma, k) = & -D^i D^j \gamma_{ij} + D^2(\text{tr } \gamma) - \langle \text{Ric}, \gamma \rangle \\ & + 2K \text{ tr } k - 2\langle K, k \rangle + \mathcal{Z}_H(\gamma; K), \end{aligned}$$

where $\mathcal{Z}_H$ is zeroth order in γ , arising from raising indices in the quadratic K -terms. The momentum linearization is

$$D\mathcal{M}_i(\gamma, k) = D_j k^j_i - D_i(\text{tr } k) + \mathcal{Z}_{M,i}(D\gamma; K) + \mathcal{Z}'_{M,i}(\gamma; DK).$$

Thus the metric component has second order in the Hamiltonian row and first order in the momentum row; the K component is algebraic in the first row and first order in the second.

At a background $z = (h, K)$, write

$$A_z = D\Phi_g(z) : E_r \longrightarrow F_r.$$

Three adjoints occur and are not interchangeable:

1. $A_z^\#$ is the differential formal L^2 -adjoint on smooth lapse-shift multipliers; on the balanced scale it maps

$$A_z^\# : \widehat{Z}_r \longrightarrow \widetilde{E}_r, \quad \widetilde{E}_r = H^r(S^2 T^* \Sigma) \oplus H^{r+1}(S^2 T^* \Sigma);$$

2. $A'_z : F'_r \rightarrow E'_r$ is the Banach transpose;

3. $A_z^\dagger = \mathcal{R}_{E_r}^{-1} A'_z \mathcal{R}_{F_r}$ is the Hilbert adjoint after the Riesz isomorphisms $\mathcal{R}_{E_r}$ and $\mathcal{R}_{F_r}$.

The formal expression $A_z^\#$, the Banach transpose, and a Sobolev-level Hilbert adjoint agree only after their declared density and order identifications. No Sobolev-dependent Hilbert adjoint is used to select the right inverse.

The formal adjoint acts on a lapse–shift pair (N, X) . Its kernel is the vacuum KID system in the frozen $K = +\frac{1}{2}\mathcal{L}_n h$ convention:

$$\mathcal{L}_X h_{ij} + 2NK_{ij} = 0, \quad (32)$$

$$D_i D_j N + \mathcal{L}_X K_{ij} - N(R_{ij} + K K_{ij} - 2K_i^k K_{kj} - \Lambda h_{ij}) = 0. \quad (33)$$

Equations (32) and (33) are the initial-data restrictions of a spacetime Killing field $Y = X + Nn$. Conversely, under the compact-CMC hypotheses used here, an adjoint kernel element develops to such a Killing field [8, 24].

24.1 Equal-row symbol and one fixed balanced normal operator

Fix once and for all a smooth positive auxiliary metric h_* on Σ and the associated density and bundle metrics. Let $\Lambda_{*,S^2} = (1 + \Delta_{*,S^2})^{1/2}$ be its positive spectral operator on symmetric two-tensors. Pointwise unitary transport U_z from the geometric L_z^2 variation pairing to the fixed h_* pairing defines

$$\mathcal{B}_{z,*} = U_z^{-1} \begin{pmatrix} \Lambda_{*,S^2}^{-2} & 0 \\ 0 & I \end{pmatrix} U_z : \tilde{E}_r \longrightarrow E_r. \quad (34)$$

This is a positive, self-adjoint, injective classical pseudodifferential operator of component orders $(-2, 0)$. Its construction depends smoothly and tamely on z through the zeroth-order transport U_z , but it is independent of the Sobolev index. It therefore defines one operator on smooth sections whose restrictions commute with all level inclusions.

The raw differential composition $A_z A_z^\#$ has order table

	N	X
$\mathcal{H}$	4	3
$\mathcal{M}$	3	2.

Its natural Douglis–Nirenberg weights are

$$s_{\mathcal{H}} = 0, \quad s_{\mathcal{M}} = -1, \quad t_N = 4, \quad t_X = 3.$$

Accordingly it maps $H^{r+4}(N) \oplus H^{r+3}(X)$ naturally to $H^r(\mathcal{H}) \oplus H^{r+1}(\mathcal{M})$, not onto the equal-row space F_r . No inverse of that raw composition on F_r is claimed. The load-bearing normal operator is instead the single, row-balanced pseudodifferential expression

$$L_z = A_z \mathcal{B}_{z,*} A_z^\# : \hat{Z}_r \longrightarrow F_r. \quad (35)$$

The declared lapse–Hamiltonian-density and shift–momentum-density pairings identify the multiplier and equation-density bundles for this purpose; every self-adjointness statement below is made after that fixed geometric identification. The same expression L_z acts on every nested level. The balancer is part of this definition; it is neither an estimate-only relabeling nor an r -dependent Moore–Penrose choice.

For the inherited equal-row target, assign column orders 2 to γ and 1 to k , and row orders 0 to both equations. The resulting principal symbol is

$$\begin{aligned} h_2(\zeta)\gamma &= -\zeta^i \zeta^j \gamma_{ij} + |\zeta|^2 \operatorname{tr} \gamma, \\ m_1(\zeta)k &= \zeta_j k^j_i - \zeta_i \operatorname{tr} k, \\ a_{z,0}(\zeta)(\gamma, k) &= (h_2(\zeta)\gamma, m_1(\zeta)k). \end{aligned}$$

The first-order γ -term in the momentum row and the algebraic k -term in the Hamiltonian row are strictly below these selected principal orders. Given a scalar and a covector at $\zeta \neq 0$, the trace/transverse part of γ realizes the scalar and longitudinal components of k realize the covector. Hence $a_{z,0}(\zeta)$ is surjective. Equivalently, its adjoint is injective; its two diagonal entries include

$$N \mapsto \zeta_i \zeta_j N - |\zeta|^2 h_{ij} N, \quad X \mapsto \zeta_i X_j - h_{ij} \zeta \cdot X.$$

Trace and contraction force $N = X = 0$.

The order-two principal symbol of the balanced normal operator is

$$\sigma_2(L_z)(\zeta) = \begin{pmatrix} h_2(\zeta) |\zeta|_*^{-2} h_2(\zeta)^* & 0 \\ 0 & m_1(\zeta) m_1(\zeta)^* \end{pmatrix}. \quad (36)$$

The off-diagonal terms have order at most one. Surjectivity of the two displayed principal rows gives

$$\langle u, \sigma_2(L_z)(\zeta) u \rangle \geq c_z |\zeta|_*^2 (|u_N|^2 + |u_X|^2), \quad \zeta \neq 0,$$

uniformly after the chart is shrunk. Thus L_z is a classical elliptic pseudodifferential operator of order two on the complete equation bundle, with unbounded L_z^2 domain $H^2(\Sigma) \oplus H^2(T\Sigma)$. On compact Σ , the elliptic parametrix theorem gives Fredholmness and elliptic regularity [19]. The formal adjoint relation and the self-adjoint balancer make L_z self-adjoint, hence its index is zero.

Finally,

$$\langle u, L_z u \rangle_{L_z^2} = \langle A_z^\# u, \mathcal{B}_{z,*} A_z^\# u \rangle_{L_z^2}. \quad (37)$$

Positivity and injectivity of $\mathcal{B}_{z,*}$ give $\ker L_z = \ker A_z^\#$, the KID space. At a no-KID point the kernel is zero, so index zero makes the fixed balanced operator $L_z : \widehat{Z}_r \rightarrow F_r$ bijective at every level by elliptic regularity.

25. An occupied vacuum no-KID chart

The regular theorem does not rest on the word “generic.” We now construct one point.

Example 25.1 (Closed-hyperbolic vacuum data). Let

$$\Sigma = \mathbb{H}^3 / \Gamma$$

for any torsion-free cocompact lattice $\Gamma < \text{Isom}^+(\mathbb{H}^3)$, and let h be its sectional-curvature -1 metric [25]. Set

$$\Lambda = 0, \quad K_{ij} = h_{ij}.$$

Then $K = 3$, $K_{ij} K^{ij} = 3$, and $R = -6$. Therefore

$$R + K^2 - K_{ij} K^{ij} = -6 + 9 - 3 = 0.$$

Since $Dh = 0$,

$$D_j K^j_i - D_i K = 0.$$

The corresponding canonical momentum is

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa} (h^{ij} - 3h^{ij}) = -\frac{\sqrt{h}}{\kappa} h^{ij}. \quad (38)$$

Thus $z_{\text{hyp}} = (h, \pi) \in \mathcal{C}_{\text{tot},g}$, and the mean curvature is the constant 3.

Theorem 25.2 (No KIDs at the occupied point). $\ker D\Phi_g(z_{\text{hyp}})^* = 0$.

Proof. At $K = h$, the first KID equation is

$$\mathcal{L}_X h = -2Nh.$$

Thus X is conformal Killing with conformal factor $\sigma = -N$: $\mathcal{L}_X h = 2\sigma h$. On a three-dimensional constant-sectional-curvature -1 manifold, the standard differentiated conformal-Killing identity gives

$$D_i D_j \sigma = \sigma h_{ij}.$$

Taking the trace,

$$D^2 \sigma = 3\sigma.$$

Multiplying by σ , integrating over compact Σ , and integrating by parts gives

$$-\int_{\Sigma} |D\sigma|^2 d\mu_h = 3 \int_{\Sigma} \sigma^2 d\mu_h.$$

Both sides can agree only when $\sigma = 0$, hence $N = 0$. The first KID equation then reduces to $\mathcal{L}_X h = 0$. The Bochner identity for a Killing field gives

$$\int_{\Sigma} |DX|^2 d\mu_h = \int_{\Sigma} \text{Ric}(X, X) d\mu_h = -2 \int_{\Sigma} |X|^2 d\mu_h,$$

so $X = 0$. Therefore the adjoint kernel is trivial. The second KID equation is then automatically satisfied only by the zero pair. $\square$

Remark 25.3 (Development). The data are realized by the quotient of the Milne-type vacuum metric

$$g = -dt^2 + t^2 h, \quad t > 0,$$

whose $t = 1$ slice has $K = h$ in the frozen convention. Its maximal globally hyperbolic development supplies the licensed vacuum history class. The quotient by Γ removes the ambient continuous Lorentz symmetries that would otherwise induce KIDs on the compact slice.

26. Vacuum split regularity (W3.4a)

At z_{hyp} , ellipticity of the fixed balanced normal operator and theorem 25.2 remove the kernel term from the order-two pseudodifferential estimate. For $q \geq q_0 > \frac{3}{2} + 4$,

$$\|u\|_{\widehat{Z}_q} \leq C_q \|L_{z_{\text{hyp}}} u\|_{F_q}. \quad (39)$$

Define the Green operator and constraint right inverse by

$$G_z = L_z^{-1} : F_r \longrightarrow \widehat{Z}_r, \quad Q_z = \mathcal{B}_{z,*} A_z^{\#} G_z : F_r \longrightarrow E_r. \quad (40)$$

Then

$$A_z Q_z = L_z G_z = \text{id}_{F_r}.$$

Lemma 26.1 (Parameter control and level compatibility). *After shrinking to a neighborhood $\mathcal{U}_{\text{hyp}}$, the following hold:*

1. the same balanced operator $L_z : \widehat{Z}_r \rightarrow F_r$ is invertible for every $z \in \mathcal{U}_{\text{hyp}}$;
2. with $q_0 > \frac{3}{2} + 4$ fixed and coefficient loss $\mu = 2$, its inverse G_z and Q_z obey, for $q \geq q_0$,

$$\|G_z f\|_{\widehat{Z}_q} \leq C_q \left(\|f\|_{F_q} + (1 + \|z\|_{E_{q+\mu}}) \|f\|_{F_{q_0}} \right), \quad (41)$$

$$\|Q_z f\|_{E_q} \leq C_q \left(\|f\|_{F_q} + (1 + \|z\|_{E_{q+\mu}}) \|f\|_{F_{q_0}} \right); \quad (42)$$

3. every parameter derivative is controlled. Explicitly, for $m \geq 1$,

$$D^m G_z[\dot{z}_1, \dots, \dot{z}_m] = \sum_{\substack{(B_1, \dots, B_k) \\ \text{ordered partition of } \{1, \dots, m\}}} (-1)^k G_z(D^{|B_1|} L_z[\dot{z}_{B_1}]) G_z \cdots (D^{|B_k|} L_z[\dot{z}_{B_k}]) G_z, \quad (43)$$

and, with $u_0 = f$, the resulting estimate has the explicit one-high-norm form

$$\|D^m G_z[\dot{z}_1, \dots, \dot{z}_m] f\|_{\widehat{Z}_q} \leq C_{q,m} (1 + \|z\|_{E_{q+\mu}}) \sum_{j=0}^m \|u_j\|_{\mathcal{X}_j(q)} \prod_{k \neq j} \|u_k\|_{\mathcal{X}_k(q_0+\mu)},$$

where $\mathcal{X}_0(q) = F_q$ and $\mathcal{X}_j(q) = E_{q+\mu}$ for $j \geq 1$; the same bound holds for every derivative of $Q_z = \mathcal{B}_{z,*} A_z^\# G_z$;

4. if $f \in F_{r+1}$, the unique distributional solutions satisfy

$$\iota_{\widehat{Z}} G_z^{(r+1)} f = G_z^{(r)} \iota_F f, \quad \iota_E Q_z^{(r+1)} f = Q_z^{(r)} \iota_F f.$$

These are restrictions of one operator G_z solving the same pseudodifferential equation $L_z u = f$, not minimum-norm solutions for changing Hilbert products; and

5. the Legendre differential carries this splitting isomorphically from (h, K) to (h, π) .

Proof. The uniform symbol inequality equation (36) and the balanced parameter-elliptic estimate give

$$\|u\|_{\widehat{Z}_q} \leq C_q (\|L_z u\|_{F_q} + \|u\|_{\widehat{Z}_{q_0}})$$

with constants locally uniform in the $E_{q+\mu}$ coefficient norm. The no-KID energy identity removes the low-norm kernel term at the occupied point. A Neumann argument for

$$L_z = L_{z_{\text{hyp}}} + (L_z - L_{z_{\text{hyp}}}) : \widehat{Z}_{q_0} \longrightarrow F_{q_0},$$

followed by the uniform elliptic estimate, preserves invertibility on one neighborhood and proves equation (41) at all higher levels.

All z -dependence in $A_z \mathcal{B}_{z,*} A_z^\#$ occurs in A_z , $A_z^\#$, and the zeroth-order transports $U_z^{\pm 1}$. The constraint coefficients contain at most two derivatives of the metric and one of K ; the fixed Λ_{*,S^2}^{-2} contributes no coefficient loss. Relative to E_q , the largest excess is therefore $\mu = 2$. Pseudodifferential composition, Sobolev multiplication, and Moser commutator estimates give the stated one-high-norm bounds for $D^j L_z$. Repeated differentiation of $L_z G_z = 1$ gives equation (43); induction with the uniform elliptic estimate proves the displayed multilinear tame bound. Composing with the differential formal adjoint $A_z^\#$ and the fixed row balancer proves equation (42) and the corresponding estimates for all derivatives of Q_z .

For level compatibility, let $f \in F_{r+1}$. Both restrictions solve the identical distributional equation $L_z u = f$. Injectivity of L_z makes the multiplier solution unique, and elliptic regularity gives $u \in \widehat{Z}_{r+1}$. Applying the same $\mathcal{B}_{z,*} A_z^\#$ proves compatibility of Q_z . No Sobolev-dependent orthogonal complement or Hilbert adjoint enters this argument. The Legendre map and its differential are smooth-tame bundle isomorphisms on the nondegenerate metric carrier, so they preserve kernels, complements, and right-inverse identities. This is the compact no-boundary specialization of the conventional constraint mapping-property framework [17, 24]; the unequal row weights of the raw $A_z A_z^\#$ composition are retained in the audit, while $\mathcal{B}_{z,*}$ converts the actual normal problem to the inherited equal-row target. The compact pseudodifferential estimates and coefficient loss used here are written explicitly rather than imported. $\square$

Theorem 26.2 (Occupied vacuum split zero locus). *There is an open neighborhood $\mathcal{U}_{\text{hyp}}$ of z_{hyp} such that:*

1. the KID space is zero at every constraint point in a possibly smaller neighborhood;

2. $D\Phi_g$ is split surjective with tame right inverse equation (40);

3.

$$\mathcal{C}_{\text{tot}g}^{\text{reg}} \cap \mathcal{U}_{\text{hyp}}$$

is a split tame Fréchet submanifold; and

4. its Sobolev level realizations are split Hilbert submanifolds.

Proof. Lemma 26.1 supplies a bounded right inverse at every level and a single compatible inverse-limit complement

$$E_r = \ker A_z \oplus \text{im } Q_z.$$

Indeed $v = (v - Q_z A_z v) + Q_z A_z v$, and $\ker A_z \cap \text{im } Q_z = 0$ follows from $A_z Q_z = 1$. The parameter estimates satisfy the hypotheses of the smooth-tame submersion theorem: Φ_g^∞ is smooth tame, $D\Phi_g$ is split surjective, and Q_z is a smooth-tame family of right inverses with the fixed loss $\mu = 2$. That theorem gives one split smooth-tame zero locus. Its restriction at each sufficiently high level is the corresponding split Hilbert zero locus, and the level-compatible charts agree on inclusions. This is the no-KID case of the Fischer–Marsden constraint-manifold theorem with the occupant, operator scales, and right inverse made explicit [8, 24]. The genericity theorem [9] confirms abundance in its stated regimes but is not used to establish occupancy. $\square$

27. Massive scalar generalized-KID gate

Certificate 27.1 (ScalarReg3). For a scalar-branch point $z = (h, \pi; \phi, p_\phi)$, set

$$\text{ScalarReg3}(z) \iff \begin{cases} \Phi_{g+\phi}(z) = 0, \\ \ker D\Phi_{g+\phi}(z)^* = 0, \\ \text{im } D\Phi_{g+\phi}(z) \text{ is closed,} \\ D\Phi_{g+\phi}(z) \text{ has a compatible tame right inverse,} \\ \text{the pointed source lift is exact,} \\ \text{the Einstein–Klein–Gordon history is licensed.} \end{cases}$$

The coupled adjoint adds matter terms to equations (32) and (33) and the condition that the spacetime symmetry preserve the scalar:

$$\mathcal{L}_X \phi + N n^\mu \nabla_\mu \phi = 0,$$

together with the corresponding momentum compatibility. These are generalized KIDs. Their absence must be proved at a specific coupled point. The 1981 scalar result and its erratum are recorded as analytic guidance, not as an occupant [1, 2].

Proposition 27.2 (Exact scalar status). *The scalar branch is PROVED for off-shell algebra and CONDITIONAL for regular reduction and finite refoliation. No statement in the vacuum occupancy proof changes that status.*

28. Direct characteristic equality (W3.4b)

Fix a full regular point z . Write

$$T_z \mathcal{P}_b^\infty = \ker D\Phi_b(z) \oplus \text{im } Q_z.$$

For $e = (N, \xi)$ in the descriptor dual, define

$$B_z(e) = X_{C_b(e)}(z).$$

The Hamiltonian identity is

$$\Omega_b(B_z(e), v) = -dC_b(e)[v] = -\langle D\Phi_b(z)v, e \rangle. \quad (44)$$

If $v \in T_z\mathcal{C}_{\text{tot}b} = \ker D\Phi_b(z)$, the right side vanishes. Hence

$$\text{im } B_z \subset \ker(\Omega_b|_{T_z\mathcal{C}_{\text{tot}b}}).$$

For the reverse inclusion, take

$$w \in \ker(\Omega_b|_{T_z\mathcal{C}_{\text{tot}b}}).$$

Put $\alpha = \Omega_b^b w$. Since α vanishes on $\ker A_z = T_z\mathcal{C}_{\text{tot}b}$, the right inverse defines one distributional multiplier

$$\lambda(f) := \alpha(Q_z f), \quad f \in F_r.$$

For every smooth v ,

$$\alpha(v) = \lambda(A_z v), \quad \text{hence} \quad \alpha = A'_z \lambda. \quad (45)$$

This is a single functional on the constraint target, not a separately selected Riesz representative at each Hilbert level.

Under the fixed density and order-reduction identifications, equation (45) is the formal-adjoint equation

$$A_z^\# \lambda = \mathcal{J}_z(\Omega_b^b w),$$

where $\mathcal{J}_z$ is the declared density/order-reduction identification from the smooth canonical cotangent density to the formal-adjoint target. Its right-hand side is smooth because w is a smooth tangent vector. Distributional elliptic regularity for the overdetermined elliptic operator $A_z^\#$, followed by bootstrapping through the smooth coefficients, makes λ a smooth lapse-shift descriptor e . Therefore

$$\Omega_b(w, v) = \langle e, A_z v \rangle = dC_b(e)[v] = -\Omega_b(B_z(e), v).$$

Weak nondegeneracy of Ω_b gives $w = -B_z(e)$.

Theorem W3.4 (Full characteristic equality). At every full regular point on a passing branch,

$$\boxed{\ker(\Omega_b|_{\mathcal{C}_{\text{tot}b}}) = \text{im } \mathfrak{R}_b.} \quad (46)$$

On the occupied vacuum chart this is unconditional. On the scalar branch it holds at every populated ScalarReg3 point.

Proof. The first inclusion is equation (44). The reverse inclusion is the single distributional factorization equation (45), followed by elliptic regularity and weak nondegeneracy. No weak double-annihilator identity and no unproved cross-level family of Riesz representatives is used. $\square$

29. Exact WDW-2 compatibility on the two constraint surfaces

The accepted WDW-2 theorem is imported on its own base:

$$\ker(\Omega_b|_{\mathcal{C}_{\text{si}b}}) = \text{im } X_{\mathcal{D}} = \ker d\pi_{\text{prop}}, \quad (47)$$

where each tangent space is a tangent space to $\mathcal{C}_{\text{si}b}$. It is not derived by setting the lapse to zero in a theorem on $\mathcal{C}_{\text{tot}b}$.

Proposition 29.1 (Intersection compatibility). *At $z \in \mathcal{C}_{\text{tot}b}$ on the common regular branch,*

$$\text{im } X_{\mathcal{D}}|_z = \ker d\pi_{\text{prop}}|_z \subset \ker(\Omega_b|_{T_z\mathcal{C}_{\text{tot}b}}). \quad (48)$$

Proof. The equality is the accepted WDW-2 orbit/vertical theorem restricted pointwise from $\mathcal{C}_{\text{si}b}$ to its subset $\mathcal{C}_{\text{tot}b}$. For the inclusion, the first-class spatial-normal and spatial-spatial brackets make $X_{\mathcal{D}[\xi]}$ tangent to $\mathcal{C}_{\text{tot}b}$. For $v \in T_z\mathcal{C}_{\text{tot}b}$,

$$\Omega_b(X_{\mathcal{D}[\xi]}, v) = -d\mathcal{D}[\xi](v) = 0.$$

Thus the tangential characteristic directions from WDW-2 sit inside the full characteristic distribution without identifying $T_z\mathcal{C}_{\text{si}b}$ with $T_z\mathcal{C}_{\text{tot}b}$. $\square$

30. KID singular strata

At a KID point, $\ker D\Phi^* \neq 0$, so $D\Phi$ is not surjective and the regular submersion theorem fails. Each nonzero KID gives:

1. isotropy of the history-embedding groupoid;
2. a degeneracy direction in the constraint-map normal operator;
3. a quadratic Taub condition on integrable linearized solutions; and
4. locally conical or higher orbit-type structure rather than a regular manifold chart.

Fischer–Marsden–Moncrief prove the smooth no-Killing case and the conical behavior near the one-Killing case in the compact-CMC setting [8]. WDW-3 uses that theorem to type the singular boundary, not to delete it.

Proposition 30.1 (Singular retention). *The global WDW-3 groupoid object contains KID points, their isotropy groups, and their second-order obstruction data. Only the local manifold, right-inverse, and characteristic quotient assertions are restricted to the regular locus.*

Proof. The object-space definition in definition 15.1 does not quotient out isotropy or discard nonregular objects. Regularity is a predicate controlling which local theorems may be invoked. Removing KID points would change the global target and erase standing-bearing symmetry data, contrary to both the conventional stratification and the WDW-2 groupoid import. $\square$

Part VI

History-embedding groupoid and finite refoliation

31. Augmented object space

For branch b , let $\text{Hist}_b^{\text{lic}}$ be the licensed reconstructed history class. Define

$$\widetilde{\text{HistEmb}}_b = \coprod_a \left\{ (a; \gamma, i, z, \widehat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R}) : \begin{array}{l} \gamma \in \text{Hist}_b^{\text{lic}}, \\ i : \Sigma \hookrightarrow M_\gamma \text{ admissible Cauchy,} \\ z = \text{Eval}_i(\gamma) \in \mathcal{C}_{\text{tot}b} \end{array} \right\}.$$

Here a is the WDW-2 presentation-cover index and $\mathcal{R}$ is the complete relational-record profile. The cover index can change by augmentation. The history, source, attachment, witness, branch, boundary certificate, and relational profile cannot.

Definition 31.1 (Evaluation on objects). $\text{Eval}_i(\gamma)$ is the canonical Cauchy datum obtained by:

1. pulling back g_γ to $h = i^*g_\gamma$;
2. forming $K = \frac{1}{2}\mathcal{L}_n h$ with the frozen future normal;
3. applying the WDW-1 Legendre map to obtain π ;
4. pulling back the canonical matter field and normal momentum when present; and
5. retaining the source, attachment, witness, boundary, and branch coordinates.

The Gauss–Codazzi projections of the covariant field equations imply $\text{Eval}_i(\gamma) \in \mathcal{C}_{\text{tot}b}$. Conversely, on a licensed regular vacuum datum, Choquet–Bruhat–Geroch development and the WDW-1 reconstruction produce a history unique up to the declared isometry relation.

32. Arrow space and structure maps

Definition 32.1 (Proper history-embedding arrow). An arrow is

$$\alpha = (a_1, i_1; a_0, i_0; \gamma; \chi)$$

where:

1. i_0, i_1 are admissible cooriented Cauchy embeddings in the same licensed history γ ;
2. $\chi : i_0(\Sigma) \rightarrow i_1(\Sigma)$ is the induced proper spacetime identification;
3. χ preserves orientation, time orientation, branch, source fibre, attachment, witness, boundary profile, and $\mathcal{R}$;
4. the associated spatial change satisfies the WDW-2 proper-arrow guard; and
5. every boundary improvement has zero physical charge on the occupied branch.

Source and target are

$$s(\alpha) = (a_0; \gamma, i_0, \text{Eval}_{i_0} \gamma, \hat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R}),$$

$$t(\alpha) = (a_1; \gamma, i_1, \text{Eval}_{i_1} \gamma, \hat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R}).$$

The unit at $u = (a; \gamma, i, \dots)$ is

$$1_u = (a, i; a, i; \gamma; \text{id}_{i(\Sigma)}).$$

The inverse is

$$(a_1, i_1; a_0, i_0; \gamma; \chi)^{-1} = (a_0, i_0; a_1, i_1; \gamma; \chi^{-1}),$$

and composable arrows obey

$$(a_2, i_2; a_1, i_1; \gamma; \chi_{21}) \circ (a_1, i_1; a_0, i_0; \gamma; \chi_{10}) = (a_2, i_2; a_0, i_0; \gamma; \chi_{21} \circ \chi_{10}).$$

Theorem 32.2 (History-embedding groupoid). *The preceding objects and arrows form a groupoid*

$$\mathbb{E}_b \rightrightarrows \widetilde{\text{HistEmb}_b}.$$

Restriction to smaller presentation charts and augmentation by WDW-2 cover arrows are groupoid functors. Every guard is stable under unit, inverse, restriction, and composition.

Proof. Source and target match by construction. Identity and inverse laws follow from those of the induced spacetime identifications. Composition is associative because composition of diffeomorphisms is associative. Preserved coordinates remain equal under composition. Zero boundary charge and properness form a subgroupoid predicate on the declared compact core; a composition involving a charge-bearing, source-changing, large, or target-changing map is therefore not an arrow of $\mathbb{E}_b$. WDW-2 cover augmentation changes only a and commutes with evaluation. These facts verify every structure and guard map rather than assuming a quotient from an untyped pair of embeddings. $\square$

33. The guarded canonical characteristic groupoid

Fix a full-regular canonical chart $U \subset \mathcal{C}_{\text{tot}_b}^{\text{reg}}$. By W3.4 its split characteristic distribution is

$$\mathcal{K}_b = \ker(\Omega_b|_{T\mathcal{C}_{\text{tot}_b}}) = \text{im } \rho_{\text{con}}.$$

At this stage no infinite-dimensional Frobenius theorem is invoked. The target is first constructed from the guarded paths that are actually tangent to $\mathcal{K}_b$. W3.6 subsequently proves, on the occupied chart, that the reachable set is a smooth-tame characteristic plaque.

Definition 33.1 (Guarded local characteristic groupoid). The objects of $\mathbb{C}_{b,U}^{\text{char}}$ are

$$(a; z, \gamma, \hat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R})$$

with $z \in U$, $\text{Recon}(z) \simeq \gamma$, and all inherited WDW-2 presentation and standing coordinates. An arrow is a characteristic-path class

$$[c] : z_0 \longrightarrow z_1, \quad \dot{c}(t) \in \mathcal{K}_b(c(t)),$$

where c stays in one regular plaque and preserves branch, history, source, attachment, witness, boundary, and relational coordinates. Two paths are equivalent exactly when they are homotopic through guarded characteristic paths relative to both endpoints. KID, charged, large, source-changing, relationally changing, and target-changing routes are not arrows of this local groupoid.

The structure maps are

$$\begin{aligned} s[c] &= c(0), & t[c] &= c(1), & 1_z &= [\text{const}_z], & [c]^{-1} &= [c^{-1}], \\ [c_2] \circ [c_1] &= [c_2 * c_1], \end{aligned}$$

where c^{-1} is path reversal and $c_2 * c_1$ is concatenation with the standard smooth reparametrization. Relative characteristic homotopy is stable under these operations, so the maps are well defined. WDW-2 pure tangential arrows embed through

$$\xi \longmapsto X_{\mathcal{D}[\xi]} \in \mathcal{K}_b$$

and proposition 29.1. Restriction to a smaller regular chart and WDW-2 cover augmentation are functorial.

Proposition 33.2 (Target construction and universal property). $\mathbb{C}_{b,U}^{\text{char}}$ is a guarded characteristic path groupoid. On the occupied chart W3.6 identifies it with the local monodromy groupoid of the constructed characteristic plaque. Every map on its object space that is constant along proper characteristic paths factors uniquely through the orbit projection

$$q_{\text{char}} : \text{Ob}(\mathbb{C}_{b,U}^{\text{char}}) \longrightarrow \text{Ob}(\mathbb{C}_{b,U}^{\text{char}}) / \mathbb{C}_{b,U}^{\text{char}}.$$

Proof. Reversal and concatenation preserve tangency to $\mathcal{K}_b$, and relative characteristic homotopy is a congruence for those operations. The guards are equality predicates on retained coordinates and are stable under units, reversal, concatenation, and homotopy. If f is constant along every arrow, define $\bar{f}([z]) = f(z)$; arrow constancy makes this independent of the representative, and surjectivity of q_{char} gives uniqueness. $\square$

34. Infinitesimal comparison and direct path evaluation

34.1 W3.5a: differentiated object evaluation

Let i_ϵ be a smooth path of embeddings with

$$\left. \frac{d}{d\epsilon} \right|_0 i_\epsilon = i_* \xi + Nn.$$

Differentiating the induced metric gives

$$\delta h_{ij} = \mathcal{L}_\xi h_{ij} + 2NK_{ij}.$$

Using the Legendre map,

$$2NK_{ij} = \frac{4\kappa N}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right),$$

so

$$\delta h_{ij} = \frac{\delta C(\xi, N)}{\delta \pi^{ij}}.$$

Differentiating K , using the Ricci evolution equation, and translating back to π gives

$$\delta \pi^{ij} = -\frac{\delta C(\xi, N)}{\delta h_{ij}}.$$

On the scalar branch,

$$\delta \phi = \mathcal{L}_\xi \phi + \frac{N p_\phi}{\sqrt{h}}, \quad \delta p_\phi = -\frac{\delta C(\xi, N)}{\delta \phi}.$$

These are precisely the Hamiltonian equations of $C(\xi, N)$.

Proposition 34.1 (Differentiated object evaluation). *At every licensed smooth history object,*

$$\boxed{d \text{Eval}_i(v_{\text{emb}}(\xi, N)) = X_{\mathcal{D}[\xi] + H_\perp[N]}.} \quad (49)$$

Proof. The metric, momentum, scalar, and scalar-momentum components are the preceding direct variations. Source and attachment derivatives vanish because the arrow lies in one fixed source fibre. The boundary and relational coordinates are fixed by the proper-arrow definition. Hence every component of the augmented evaluation map has the displayed differential. $\square$

34.2 W3.5b: coefficient-sensitive on-shell morphism

Over the object evaluation map, define

$$\Psi_i(e) = -[C_b(e)]_{\text{Eval}_i(\gamma)}.$$

For $f \in C^\infty(U)$, the base-module rule is

$$\Psi_i((\text{Eval}^* f)e) = f \Psi_i(e).$$

The differentiated evaluation proposition gives both commuting identities

$$d \text{Eval} \circ \rho_{\text{src}} = \rho_{\text{con}} \circ \Psi = -X_{C_b(\cdot)} \quad (50)$$

and

$$\rho_{\text{src}}(e)(\text{Eval}^* f) = \text{Eval}^*(\rho_{\text{con}}(\Psi e)f). \quad (51)$$

Combining equations (26) and (51) with the two Leibniz rules yields

$$\Psi([e_1, (\text{Eval}^* f)e_2]_{\text{src}}) = [\Psi e_1, f \Psi e_2]_{\text{con}}. \quad (52)$$

Thus Ψ is promoted here—and only here—from the fixed-frame map Ψ_0 of W3.3 to the complete source-to-conormal Lie–Rinehart/algebroid morphism.

Proof of the coefficient identities. Since $\rho_{\text{src}}(e) = -v_{\text{emb}}(e)$, equation (49) gives

$$d \text{Eval}(\rho_{\text{src}}(e)) = -X_{C_b(e)} = \rho_{\text{con}}(-[C_b(e)]) = \rho_{\text{con}}(\Psi e),$$

which is equation (50). Applying both sides to f proves equation (51). The source Leibniz term is therefore carried to the canonical Leibniz term, while the frame terms agree by W3.3c. This proves equation (52) without using any finite functor or downstream theorem. $\square$

34.3 W3.5c: direct path and homotopy functor

The forward path map requires no Lie-II theorem. For $r \geq q_0$, use the shifted embedding and canonical levels

$$\mathcal{E}_r = \text{Emb}^{r+3}(\Sigma, M_\gamma), \quad \mathcal{P}_r = H^{r+2}(S^2 T^* \Sigma) \oplus H^{r+1}(S^2 T \Sigma \otimes \text{Dens } \Sigma). \quad (53)$$

The fixed three-index embedding shift records one derivative in pullback and a second derivative in the unit normal and second fundamental form. Define

$$\mathcal{P}_r^1(\mathcal{E}) = C^0([0, 1], \mathcal{E}_{r+1}) \cap C^1([0, 1], \mathcal{E}_r),$$

and use the analogous two-parameter space, with one continuous and one weak derivative in each path parameter, for guarded homotopies. Pullback, normal formation, the WDW-1 Legendre map, and retained-coordinate identities map these spaces continuously into the corresponding $\mathcal{P}_r$ path and homotopy spaces. The complete smooth-tame component estimates are proved in W3.6a; only this forward, fixed-shift mapping property is used here.

Let $\Pi_1^{\text{guard}}(\mathcal{O}_{\text{emb}})$ be guarded embedding paths modulo guarded homotopy relative endpoints. Define directly

$$\text{Eval}^{\text{path}} : \Pi_1^{\text{guard}}(\mathcal{O}_{\text{emb}}) \longrightarrow \mathbb{C}_{b,U}^{\text{char}}, \quad [i_\bullet] \longmapsto [\text{Eval}_{i_\bullet}(\gamma)]. \quad (54)$$

Lemma 34.2 (Path-functor laws). *Eval^{path} is independent of path and homotopy representatives and preserves source, target, unit, inverse, composition, chart restriction, presentation augmentation, and every guard.*

Proof. A differentiable embedding path has a unique tangential-normal decomposition $\dot{i} = i_* X + Nn = v_{\text{emb}}(X, N)$. By equation (49), its image velocity is $X_{C_b(X, N)}$, hence lies in $\mathcal{K}_b$. Applying the same calculation to each path in a guarded two-parameter homotopy shows that its image is a guarded characteristic homotopy. Therefore equation (54) is independent of both representatives.

Endpoint evaluation gives preservation of source and target. A constant embedding path evaluates to a constant canonical path; time reversal commutes with evaluation; and pointwise evaluation commutes with the standard smoothing of concatenation. These are the unit, inverse, and composition laws. Restriction is literal restriction of the same path. WDW-2 augmentation changes only the cover coordinate and commutes with object evaluation. Finally, branch, history, source, attachment, witness, boundary, and relational coordinates are held fixed along a source path, so their identity maps preserve every target guard. Each assertion holds on the compatible spaces following equation (53); the level inclusions commute with pullback and evaluation before passage to smooth points. $\square$

On the licensed regular vacuum class, reconstruction sends a target object to its maximal globally hyperbolic development and distinguished embedding class. WDW-1 reconstruction and Choquet–Bruhat–Geroch uniqueness make this object map inverse to evaluation up to the isometry arrows already retained. For a scalar point the same statement is conditional on the history clause of ScalarReg3.

Theorem W3.5 (Constructed target and direct on-shell path evaluation). *On every licensed full-regular chart:*

1. $\mathbb{C}_{b,U}^{\text{char}}$ is the guarded local characteristic path groupoid;
2. $\Psi : \mathcal{E}_{\text{def}} \rightarrow N^* \mathcal{C}_{\text{tot}b}$ is the sign-aligned on-shell Lie-algebroid morphism completed by equation (52);
3. $\text{Eval}^{\text{path}}$ is the directly constructed functor equation (54); and
4. physical forward evaluation obeys equation (49).

The theorem is unconditional on the occupied vacuum chart and conditional on ScalarReg3 for the scalar regular quotient.

Proof. Proposition 33.2 proves item 1. W3.3c, equations (50) and (52), and proposition 34.1 prove items 2 and 4. Lemma 34.2 proves item 3 directly. Crainic–Fernandes is retained only as finite-dimensional structural comparison; no theorem from it is used to integrate this infinite-dimensional carrier [18]. $\square$

This construction closes the phase-space interface explicitly left open by the bare Σ -evolution groupoid construction [7].

35. Local finite path coherence

Let $\mathcal{U}$ be a contractible regular embedding chart in one licensed history, small enough that:

1. every embedding is admissible and has the same orientation and boundary profile;
2. evaluation lands in one full-regular canonical chart;
3. the characteristic distribution has constant split rank; and
4. the local Cauchy evolution and evaluation maps are defined;
5. the no-KID anchor is injective; and
6. the embedding chart is convex in a fixed local addition.

Lemma 35.1 (Anchor isomorphism). *At every point of $\mathcal{U}$, the conormal anchor is an isomorphism from the descriptor fibre onto the characteristic distribution.*

Proof. Surjectivity is W3.4. If $X_{C_b(e)} = 0$, weak nondegeneracy gives $dC_b(e) = A'_ze = 0$, equivalently $A_z^\#e = 0$ after the fixed density identification. Thus e is a KID, and the no-KID hypothesis gives $e = 0$. $\square$

35.1 W3.6a: smooth-tame evaluation and its differentiated inverse

The shifted tower equation (53) is not cosmetic. Its component ledger is

component	input	output	fixed loss
$i \mapsto i^*g_\gamma$	$\mathcal{E}_r$	$H^{r+2}(S^2T^*\Sigma)$	1
$i \mapsto n_i$	$\mathcal{E}_r$	$H^{r+2}(i^*TM_\gamma)$	1
$i \mapsto K_i$	$\mathcal{E}_r$	$H^{r+1}(S^2T^*\Sigma)$	2
$(h, K) \mapsto (h, \pi)$	E_r	$\mathcal{P}_r$	0
$e \mapsto v_{\text{emb}}(e)$	$\hat{Z}_{r+1} = H^{r+3} \oplus H^{r+3}$	$T\mathcal{E}_r$	1
$\mathcal{J}_{E,z}\Omega^b$	$\mathcal{K}_z \cap \mathcal{P}_{r+3}$	E_{r+1}	2
$G_z A_z \mathcal{B}_{z,*} \mathcal{J}_{E,z}\Omega^b$	$\mathcal{K}_z \cap \mathcal{P}_{r+3}$	$\hat{Z}_{r+1}$	3 overall
$d\text{Eval}_i$	$T_i\mathcal{O}_{\text{emb}} \cap \mathcal{E}_{r+2}$	$\mathcal{K}_{\text{Eval}(i)} \cap \mathcal{P}_r$	2.

In local coordinates, every derivative of pullback is a finite sum of products of derivatives of i , g_γ , and inverse induced metrics. The normal and second-fundamental-form rows add at most two derivatives. The compact Sobolev composition and Moser estimates therefore show that

$$\text{Eval} : \mathcal{E}_\infty \longrightarrow \mathcal{P}_\infty$$

is smooth tame with forward loss $\ell_{\text{Eval}} = 2$. The retained coordinates are identity or constant maps and add no loss. The inverse family below has the separately declared fixed loss $\ell_R = 3$; the extra level supplies a $\hat{Z}_{r+1}$ multiplier whose lapse and shift both lie in H^{r+3} , as required for a tangent vector to $\mathcal{E}_r = \text{Emb}^{r+3}$.

Let

$$\mathcal{D}_z^\infty = \Omega_z^b(T_z \mathcal{P}_b^\infty) \subset T_z^* \mathcal{P}_b^\infty$$

be the admitted smooth cotangent-density subbundle (the image of the weak canonical form), and let

$$\mathcal{J}_{E,z} : \mathcal{D}_z^\infty \longrightarrow E_\infty$$

be the pointwise density, bundle-musical, and Legendre identification. In canonical coordinates it is the algebraic exchange of the $(\delta h, \delta \pi)$ cotangent-density components, followed by the derivative of the $(h, \pi) \leftrightarrow (h, K)$ Legendre map. It is one smooth-tame bundle map, independent of the Sobolev index; it is not a Hilbert Riesz map. Put

$$b_z(w) = \mathcal{J}_{E,z}(\Omega_z^b w).$$

For $w \in \mathcal{K}_z$, the single multiplier constructed in equation (45) is

$$\lambda_z(w) = Q'_z \Omega_z^b w \in F'_r.$$

Because $A_z Q_z = 1$,

$$Q'_z A'_z = 1, \quad A'_z \lambda_z(w) = \Omega_z^b w.$$

The fixed formal-adjoint identification turns the latter identity into

$$A_z^\# \lambda_z(w) = b_z(w).$$

Multiplying by the fixed row balancer and applying A_z gives

$$L_z \lambda_z(w) = A_z \mathcal{B}_{z,*} b_z(w).$$

Distributional elliptic regularity for the balanced order-two operator makes $\lambda_z(w)$ a member of $\widehat{Z}_{r+1}$ when $w \in \mathcal{K}_z \cap \mathcal{P}_{r+3}$. The unique inverse of the same pseudodifferential equation on every level gives

$$\lambda_z(w) = G_z A_z \mathcal{B}_{z,*} b_z(w).$$

Thus the physical embedding descriptor is the correctly typed element

$$\boxed{e_z(w) = -\lambda_z(w) = -G_z A_z \mathcal{B}_{z,*} \mathcal{J}_{E,z}(\Omega_z^b w) \in \widehat{Z}_{r+1}.} \quad (55)$$

The equality between the distributional and pseudodifferential formulas follows from the displayed formal-adjoint equation and uniqueness of L_z ; no raw $A_z A_z^\#$ inverse and no unequal-row relabeling is introduced by notation. The estimates of lemma 26.1, the bounded component maps

$$\begin{aligned} \Omega^b : \mathcal{P}_{q+3} &\rightarrow \mathcal{D}_{z,q+1}, & \mathcal{J}_{E,z} : \mathcal{D}_{z,q+1} &\rightarrow E_{q+1}, \\ \mathcal{B}_{z,*} : E_{q+1} &\rightarrow E_{q+1}, & A_z : E_{q+1} &\rightarrow F_{q+1}, \\ G_z : F_{q+1} &\rightarrow \widehat{Z}_{q+1}, \end{aligned}$$

and the Moser estimates give

$$\|e_z(w)\|_{\widehat{Z}_{q+1}} \leq C_q \left(\|w\|_{\mathcal{P}_{q+3}} + (1 + \|z\|_{E_{q+1+\mu}}) \|w\|_{\mathcal{P}_{q_0+3}} \right), \quad \ell_R = 3, \quad \mu = 2, \quad (56)$$

and the analogous multilinear estimate for every derivative in (z, w) .

Define

$$R_i(w) = v_{\text{emb}}(e_{\text{Eval}(i)}(w)). \quad (57)$$

The sign in equation (55) is fixed by $\Omega^b X_{C(e)} = -A'_z e$. Consequently,

$$d \text{Eval}_i \circ R_i = \text{id}_{\mathcal{K}_z}, \quad R_i \circ d \text{Eval}_i = \text{id}_{T_i \mathcal{O}_{\text{emb}}}. \quad (58)$$

For the first identity, the construction gives $w = -X_{C_b(\lambda_z(w))} = X_{C_b(e_z(w))}$. For the second, if $w = d \text{Eval}_i(v_{\text{emb}}(d)) = X_{C_b(d)}$, then $\Omega^b w = -A'_z d$, so $\lambda_z(w) = Q'_z(-A'_z d) = -d$, and $e_z(w) = d$. Thus the first identity uses characteristic equality and the second uses $Q'_z A'_z = 1$, equivalently no-KID uniqueness. Equations (41), (43) and (56) make R_i a smooth-tame inverse family with the declared fixed loss.

Lemma 35.2 (Nash–Moser local evaluation equivalence). *After shrinking $\mathcal{U}$, evaluation is a smooth-tame local diffeomorphism from the embedding orbit onto a contractible characteristic plaque.*

Proof. Choose the split constraint-surface chart κ supplied by W3.4 at $z_0 = \text{Eval}(i_0)$. The tame projection required here is constructed rather than assumed. Extend the descriptor formula to $u \in T_{z_0}\mathcal{C}_{\text{tot}b}$ by

$$e_{z_0}^{\text{ext}}(u) = -G_{z_0}A_{z_0}\mathcal{B}_{z_0,*}\mathcal{J}_{E,z_0}(\Omega_{z_0}^b u),$$

and set

$$P_{\mathcal{K}}u = X_{C_b(e_{z_0}^{\text{ext}}(u))}. \quad (59)$$

All factors are smooth tame on the declared shifted tower. If $u \in \mathcal{K}_{z_0}$, equation (55) gives $P_{\mathcal{K}}u = u$; hence $P_{\mathcal{K}}^2 = P_{\mathcal{K}}$. Consequently

$$T_{z_0}\mathcal{C}_{\text{tot}b} = \mathcal{K}_{z_0} \oplus S_{z_0}, \quad S_{z_0} = \ker P_{\mathcal{K}},$$

is a proved tame direct sum. After shrinking, $P_{\mathcal{K}}|_{\mathcal{K}_z}$ is invertible; its inverse is the Neumann series of a smooth-tame small family. Thus the derivative of

$$\mathcal{F} = P_{\mathcal{K}} \circ \kappa \circ \text{Eval}$$

has the smooth-tame inverse

$$(d\mathcal{F}_i)^{-1} = R_i \circ (P_{\mathcal{K}}|_{\mathcal{K}_z})^{-1}.$$

The forward estimates above, equation (56), and the parameter estimates for R_i verify the hypotheses of Hamilton’s Nash–Moser inverse-function theorem [20]. Hence $\mathcal{F}$ is a local smooth-tame diffeomorphism.

In the split chart, the remaining S_{z_0} -coordinate of $\text{Eval} \circ \mathcal{F}^{-1}$ is a smooth-tame function of the $\mathcal{K}_{z_0}$ -coordinate. Its graph has tangent $\mathcal{K}_z$ by equation (49); it is therefore the required characteristic plaque. Shrinking the domain to a convex ball makes both the embedding orbit and its graph contractible. This proves local bijectivity and the smooth-tame inverse without a same-index Hilbert inverse-function claim. $\square$

35.2 W3.6b: local pair-groupoid and same-arrow theorem

Lemma 35.3 (Trivial local isotropy and holonomy). *On the shrunk chart, the embedding orbit and characteristic plaque have trivial isotropy, monodromy, and local characteristic holonomy. Their guarded path groupoids are pair groupoids.*

Proof. No-KID injectivity eliminates infinitesimal and sufficiently small finite isotropy. The plaque is the graph constructed in the split chart, so its normal bundle is trivialized by the fixed S_{z_0} -coordinate and a null-homotopy transports that coordinate trivially. The plaque and embedding orbit are contractible; hence every fixed-endpoint pair of paths is homotopic relative endpoints. Their guarded path groupoids therefore contain exactly one arrow for each ordered endpoint pair. $\square$

Theorem 35.4 (Local finite-refoliation coherence). *On a contractible regular embedding chart, direct and composite proper deformation paths with the same endpoints induce the same arrow in the canonical characteristic quotient and the same source, reconstruction, Dirac-observable, relational-record, and boundary maps.*

Proof. Lemma 35.2 identifies the embedding orbit with a characteristic plaque. Lemma 35.3 then proves, rather than assumes, that both local path groupoids are pair groupoids. The direct functor of W3.5 consequently descends to the original χ -arrow history groupoid and maps the unique source endpoint arrow to the unique target endpoint arrow. Every retained coordinate is constant along that arrow by the guards and by the coherence propositions below. $\square$

Nonclaim 35.5 (No global path-independence claim). The theorem does not identify arbitrary globally distinct routes. Isotropy, topology, mapping class, boundary charge, source change, and relational holonomy may distinguish same-endpoint routes outside the contractible regular chart.

36. Global residual classifier

Definition 36.1 (Exhaustive residual classes). A difference between two finite routes is assigned to the first applicable typed class:

1. proper homotopy inside one regular chart;
2. KID or other isotropy action;
3. mapping-class or large transformation;
4. nonzero boundary charge, flux, or central extension;
5. changed source, attachment, or witness;
6. changed relational record or clock-use profile;
7. changed topology, orientation, time orientation, branch, or theorem target; or
8. failure of history existence, admissibility, common domain, or regularity.

Proposition 36.2 (Exhaustion). *Every attempted comparison of two finite deformation routes is either a proper local refoliation or has a nonempty residual class in definition 36.1.*

Proof. First compare theorem targets and existence; failure gives classes 7 or 8. With a common history and target, compare source and relational coordinates, giving classes 5 or 6. Then compare boundary charge and global transformation class, giving classes 3 or 4. At fixed remaining data, nontrivial stabilizer gives class 2. If none applies, the routes are in the proper regular local groupoid and class 1 applies. The decision order is machine-recorded and prevents representative multiplicity from being mistaken for a new physical class. $\square$

Only class 1 is quotient-trivial by the occupied core theorem.

37. Source-fibre and reconstruction coherence

Let $\alpha : i_0 \rightarrow i_1$ be a proper history-embedding arrow. Since its object tuple contains one fixed $(\hat{x}, \text{Att}, W)$,

$$\overline{\text{Src}} \circ \text{Eval}_{i_1} = \overline{\text{Src}} \circ \text{Eval}_{i_0} = \hat{x}.$$

Likewise,

$$\overline{\text{Recon}}(\text{Eval}_{i_1} \gamma) \simeq \gamma \simeq \overline{\text{Recon}}(\text{Eval}_{i_0} \gamma).$$

These are equalities in the structured source quotient and natural isomorphisms in the history groupoid, not equality of bare tensor components.

Proposition 37.1 (Pointed scalar-fibre preservation). *At a populated ScalarReg3 point, finite proper refoliation preserves the exact Klein–Gordon pointed-lift fibre.*

Proof. The scalar field is one covariant history field pulled back to two embeddings. Its canonical momentum is the corresponding normal derivative density. Evaluation changes the pair by the scalar Hamiltonian equations but does not change the pointed source label, attachment, or history. The ScalarReg3 history clause supplies existence and uniqueness; no tensor-only inference is used. $\square$

38. Observable and relational coherence

For a Dirac observable O_D ,

$$\{O_D, C(\xi, N)\}|_{\mathcal{C}_{\text{tot}}} = 0,$$

so O_D is constant on each full characteristic leaf. A relational observable has the form

$$O_{A|T=\tau}[\gamma],$$

a correlation between a partial observable A and an eligible clock record T in the complete history. It need not be constant as τ varies. Refoliation invariance says that two embeddings compute the same complete correlation profile, not that the profile is constant.

Record variables retain their formation, readout, and continuation data. Boundary observables are constant only under proper zero-charge arrows; a boundary Hamiltonian may generate genuine evolution.

39. Finite refoliation theorem

Theorem W3.6 (Finite refoliation theorem). On every occupied regular branch, $\mathbb{E}_b$ locally integrates the full characteristic distribution and preserves the complete source, reconstruction, Dirac-observable, relational-record, and boundary profile. Globally, every residual is one of the classes in definition 36.1; no other difference is silently collapsed.

Proof. Local integration and endpoint coherence are theorem 35.4. Source and reconstruction coherence are the preceding identities. Dirac constancy follows from the characteristic definition, and relational-profile constancy follows from evaluation in one fixed history. Boundary preservation is an arrow guard. Exhaustion is proposition 36.2. $\square$

Part VII

Boundary and gauge classification

40. Compact-core differentiability

On a manifold with boundary, variation of the spatial functional produces

$$\delta\mathcal{D}[\xi] = \text{bulk} + 2 \int_{\partial\Sigma} n_j \xi^i \delta\pi^j_i + \int_{\partial\Sigma} \mathcal{V}_D(h, \pi, \xi; \delta h),$$

where $\mathcal{V}_D$ is the metric contribution obtained when the connection in $-2D_j\pi^j_i$ is varied. Equivalently, the positive-Lie-drag expression $\int \pi^{ij}\mathcal{L}_\xi h_{ij}$ already includes the corresponding surface term relative to the bulk momentum constraint.

The curvature part of the normal functional gives

$$\begin{aligned} \delta V[N]|_{\partial\Sigma} = & -\frac{1}{2\kappa} \int_{\partial\Sigma} \sqrt{\sigma} [N n^k (D^i \delta h_{ik} - D_k (h^{ij} \delta h_{ij})) \\ & - (n^i D^j N - h^{ij} n^k D_k N) \delta h_{ij}], \end{aligned}$$

up to the boundary-orientation convention recorded in Appendix A. A finite-boundary generator is not differentiable until the boundary conditions make this term vanish or a specific $B[\xi, N]$ cancels it.

For the occupied core,

$$\partial\Sigma = \emptyset.$$

All displayed boundary integrals are therefore empty. Consequently the bulk functionals are differentiable on $\mathcal{A}_{3b}$, the bracket calculations have no central surface term, and every compactly supported lapse-shift descriptor is proper relative to the boundary-charge test. This establishes differentiability; it does not imply boundary triviality in a different theorem target.

41. Conditional finite-boundary extension

Definition 41.1 (Boundary generator certificate). A boundary extension passes only if it records:

1. the boundary type and phase space;
2. fixed and variable boundary data;
3. allowed lapse–shift descriptors;
4. the full surface variation of the bulk generator;
5. an improvement $B[\xi, N]$ making the generator differentiable;
6. symplectic flux and integrability of the charge variation;
7. the charge, any central or field-dependent extension, and its normalization;
8. whether the arrow is proper, improper, or sector-changing; and
9. compatibility with source, attachment, and global transformation data.

Boundary class	Required improvement	Standing test	WDW-3 status
Compact, no boundary	None	All surface charges absent	OCCUPIED
Finite Dirichlet	Gibbons–Hawking/York-compatible canonical boundary term and corner data	Fixed induced geometry versus allowed boundary motion	CONDITIONAL
Finite mixed/Robin	Ensemble-specific improvement and flux control	Integrable zero charge versus physical boundary work	CONDITIONAL
Asymptotically flat	Regge–Teitelboim parity, falloff, and ADM improvement	Poincaré charge zero or nonzero	CONDITIONAL
Asymptotically AdS or other	Asymptotic counterterm and symmetry certificate	Conserved charge and possible extension	CONDITIONAL

The proper subalgebroid consists only of descriptors for which the improved charge vanishes. If a boundary symmetry has nonzero charge, its Hamiltonian direction is not in the quotient kernel even if its bulk density is a linear combination of constraints.

42. Asymptotic time and boundary evolution

For asymptotically flat data in asymptotically Cartesian coordinates, the Regge–Teitelboim improvement has the leading form

$$B_\infty[N, N^i] = \lim_{r \rightarrow \infty} \int_{S_r} dS_i \left[\frac{1}{2\kappa} \{N(\partial_j h_{ij} - \partial_i h_{jj}) - (h_{ij} - \delta_{ij} h_{kk}) \partial_j N\} + 2N^j \pi_j^i \right],$$

with the exact background subtraction, density conversion, and parity conditions fixed by the selected asymptotic phase space [13]. For $N \rightarrow 1$, $N^i \rightarrow 0$, this is the ADM energy in the frozen normalization. Thus

$$G_{\text{imp}}[1, 0]|_{\mathcal{C}_{\text{tot}}} = E_{\text{ADM}},$$

not zero in general.

Proposition 42.1 (Boundary-standing non-collapse). *No deformation with nonzero improved boundary charge is an arrow of the proper refoliation groupoid.*

Proof. Its generator remains nonzero on the bulk constraint locus and changes a standing-bearing boundary observable. Quotienting it would identify physically distinct charge sectors or boundary evolution. The arrow therefore falls in residual class 4 of definition 36.1. $\square$

Bulk/boundary distinction. “The Hamiltonian is a constraint” is a statement about the compact boundaryless bulk or a proper zero-charge descriptor. It does not erase ADM energy, quasilocal energy, an allowed boundary clock, or another populated boundary Hamiltonian.

43. Projectable descriptor audit

The reduced ADM presentation suppresses the primary lapse and shift momenta only after their extended-phase-space role has been controlled. In the extended presentation,

$$P_N = 0, \quad P_i = 0$$

are primary constraints. A projectable gauge generator has the schematic form

$$G[\epsilon] = \dot{\epsilon}^\mu P_\mu + \epsilon^\mu (\mathcal{H}_\mu + N^\rho C_{\mu\rho}{}^\nu P_\nu), \quad (60)$$

with the descriptor map depending on lapse and shift as required by the Legendre projectability condition. The precise coefficients follow the complete constraint algebra and the convention for spacetime descriptors.

Pons–Salisbury–Shepley show that arbitrary coordinate components of a spacetime vector field are not automatically projectable as canonical transformations; the projectable gauge descriptors depend on lapse and shift, and the generator acts on the full phase space [14].

Proposition 43.1 (Extended/reduced equivalence boundary). *For observables independent of the multiplier momenta, and after the descriptor map and primary constraints in equation (60) are imposed, the induced transformation of $(h, \pi; \phi, p_\phi)$ agrees with the reduced generator*

$$C(\xi, N) = \mathcal{D}[\xi] + H_\perp[N].$$

Proof. The primary terms generate the transformations of lapse and shift required to keep the Legendre map projectable. Their Poisson brackets with the reduced canonical variables vanish except through the multiplier-dependent coefficients. Substitution of the projectable descriptor relation leaves the secondary generator on the reduced variables, whose direct canonical variations were calculated in sections 17 to 19. The equivalence does not identify arbitrary spacetime vector fields with reduced gauge arrows; it applies only to the projectable class. $\square$

The WDW-3 core uses the reduced presentation because lapse and shift are multipliers and the theorem concerns the canonical history data. The machine ledger nevertheless records the primary completion and prevents the reduced formula from being advertised as a transformation of every extended variable.

Part VIII

AASC standing, clocks, and refoliation

44. Why normal deformation is constrained

The conventional chain is established before the standing consequence:

1. the lapse and shift enter the reduced canonical action as multipliers;
2. their variations impose $\mathcal{H}_\perp = 0$ and $\mathcal{H}_i = 0$;
3. the complete off-shell constraint brackets close with the metric-dependent hypersurface-deformation structure;
4. on the full regular locus, the generated directions are exactly the characteristic kernel of the pulled-back canonical form;
5. differentiated history evaluation identifies those directions with changes of Cauchy embedding; and
6. local finite proper changes preserve the complete augmented history object.

Only after those obligations are discharged does AASC determine the quotient consequence. The AASC kernel is not an optional interpretation placed over independently complete physics. Determinacy, identity, comparison, admissible transformation, standing-bearing use, and irreversible fixation presuppose admissibility, standing, reference, and irreversibility. Every determinate canonical state, reconstructed history, source fibre, record, and boundary endpoint therefore instantiates the kernel by necessity [3].

Theorem 44.1 (No new standing from proper refoliation). *Fix the complete WDW-3 target coordinates:*

$$(\gamma, \hat{x}, \text{Att}, W, \overline{\mathcal{O}}, \mathcal{R}, \text{BdyCert}, \text{branch}, \text{topology}, \text{orientation}).$$

Two regular presentations related by a proper refoliation arrow have the same admitted terminal standing. Slicing alone supplies no additional standing-bearing residue.

Proof. The finite-refoliation theorem constructs the arrow and proves preservation of every listed coordinate. The residual classifier shows that every standing-bearing difference falls outside the proper class. The remaining distinction is therefore only the representative embedding or foliation label. Under fixed-domain identity and scope-preserving comparison, representative multiplicity has no independent authority. Quotienting it is forced by faithful identity comparison, not adopted as a discretionary interpretive overlay. $\square$

This theorem does not say that nothing happens. The history can contain nontrivial curvature, matter propagation, relational records, and boundary evolution. It says that two slicings of that same complete object are not two physical histories.

45. Lapse and foliation role classifier

Object	Default compact-core role	Promotion condition
Bulk lapse N	Multiplier and normal-presentation coordinate	A declared boundary or relational structure gives the use standing
Shift N^i	Spatial-redescription multiplier	Nonzero boundary charge, large action, or target enrichment
Foliation label	Skin within one proper refoliation class	Changed history, source, boundary, clock profile, topology, or orientation
Relational clock	Standing-bearing use-relative quotient class	Must pass the physical-clock eligibility, readout, record, and continuation certificate
Boundary time	Boundary anchor/tensor coordinate	Retained when its improved Hamiltonian charge is nonzero
One selected slice	Local computational presentation	Never a global selector without independent standing-bearing authority

Lemma 45.1 (Multiplier is not a clock). *The facts that $N > 0$, that N labels a smooth Cauchy splitting, or that $N = 1$ is computationally convenient do not establish a physical clock.*

Proof. A clock requires an eligible carrier, a readout relation, a record burden, retained continuation, and a failure boundary. A lapse smearing by itself supplies none of those coordinates. It parametrizes a normal deformation of presentation. A boundary lapse or relational field can acquire clock use only after the missing coordinates are supplied. $\square$

46. No primitive preferred foliation theorem

Theorem 46.1 (No same-target foliation selector). *Within the occupied compact-core WDW-3 target, none of the following selects a primitive preferred foliation:*

1. *existence of a smooth Cauchy temporal function;*
2. *a convenient gauge condition;*
3. *one positive lapse or shift;*
4. *one WDW-2 local slice;*
5. *one numerical discretization or evolution scheme;*
6. *one semiclassical matter variable not included in the classical target; or*
7. *one boundary frame absent from the compact-core object.*

Proof. Smooth splitting proves existence of presentations, not uniqueness or physical privilege. A gauge condition and local slice are sections of a quotient presentation and have no authority outside their chart. A lapse and shift are multipliers. Numerical schemes add algorithmic data, while a semiclassical clock changes the branch and a boundary frame changes the boundary target. Any genuine selector is therefore either declared physical structure, target enrichment, representative privilege, or unsupported authority. The first two are retained as new coordinates; the latter two cannot define standing. $\square$

Corollary 46.2 (Refoliation no-selector result). *No preferred foliation can be derived by choosing a representative of the already constructed refoliation quotient. Such a choice is a section, not a theorem of new physical standing.*

47. Clock-quotient compatibility

The Physical Clock as a Quotient theorem classifies a clock through its use-relative boundary, admissibility and eligibility data, readout, record burden, retained continuation, failure boundary, and empty/singleton/plural quotient status [21]. Let

$$\mathfrak{C}_T = (B_T, \text{Adm}_T, \text{Elig}_T, \text{Read}_T, \text{Rec}_T, \text{Cont}_T, \text{Fail}_T, Q_T)$$

denote that certificate.

Theorem W3.7 (Source, observable, boundary, and clock coherence). A proper WDW-3 refoliation arrow preserves $\mathfrak{C}_T$ and its quotient status. WDW-3 neither chooses a clock nor identifies two histories with different clock-use profiles.

Proof. The history arrow preserves the complete relational-record coordinate $\mathcal{R}$, of which $\mathfrak{C}_T$ is a typed component. Evaluation on a different slice may change the representative values used to compute a correlation, but the complete correlation, readout, record, continuation, and failure data remain those of the same history. If any coordinate changes, residual class 6 applies and the comparison is not a proper refoliation. Empty, singleton, and plural quotient cardinalities are therefore invariant under the proper arrow. $\square$

Corollary 47.1 (No external bulk time). *On the compact boundaryless total constraint locus,*

$$H_{\text{can}}[N, N^i] = H_{\perp}[N] + \mathcal{D}[N^i]$$

vanishes as a function, and its regular Hamiltonian directions are characteristic. It does not generate evolution with respect to an independently standing external bulk time.

Proof. Both constraint densities vanish on $\mathcal{C}_{\text{tot}}$, and Theorem W3.4 identifies their directions with the characteristic kernel. A relational clock remains possible by Theorem W3.7, and a boundary Hamiltonian remains possible by proposition 42.1. Thus the corollary excludes only an undeclared same-scope external bulk clock. $\square$

Part IX

Rigidity comparison and capstone

48. HKT comparison without circularity

The Hojman–Kuchař–Teitelboim theorem begins from a representation of hypersurface deformations by canonical functionals of an intrinsic metric and its conjugate momentum, subject to its stated locality, differential-order, variable-content, and time-reversibility hypotheses. Under those assumptions it identifies Einsteinian geometrodynamics as the representative in the relevant class [6].

WDW-3 does not use that theorem to derive the WDW-1 Hamiltonian. The order is:

AASC gravity and Einstein-source import $\longrightarrow$ WDW-1 ADM Legendre calculation
 $\longrightarrow$ WDW-3 direct HDA calculation
 $\longrightarrow$ HKT overlap comparison.

HKT coordinate	WDW-3 overlap	Non-overlap/guard
Intrinsic metric and conjugate momentum	Vacuum branch uses exactly (h, π)	Scalar and internal branches contain additional variables
Hypersurface-deformation representation	Directly proved off shell	Not assumed as a premise for the bracket calculation
Local differential functional	WDW-1 vacuum density has the admitted local form	Nonlocal, higher-derivative, or boundary-enriched theories are outside the comparison
Time reversibility	Vacuum Einstein bulk is in the overlapping class	AASC irreversibility is a standing/fixation order, not inserted as an HKT canonical term
Compact boundaryless target	Occupied WDW-3 core	Boundary Hamiltonians require a different theorem object
Uniqueness conclusion	Agreement with the Einsteinian representative	No claim of uniqueness outside the exact HKT hypotheses

Proposition 48.1 (Noncircular HKT overlap). *The WDW-1 vacuum generator belongs to the HKT comparison class on the common compact boundaryless domain and agrees with the Einsteinian representative there. This is a consistency and rigidity comparison, not a derivation of an already imported premise.*

Proof. The variable content, local differential form, time-reversible vacuum bulk, and direct HDA representation match the displayed HKT coordinates. WDW-1 obtains the generator from the Einstein–Hilbert/ADM Legendre calculation before HKT is invoked. Therefore the conclusion checks membership and agreement; it is not used upstream to manufacture the generator. $\square$

49. The W3.1–W3.8 dependency chain

Gate	Endpoint	Branch status
W3.1	Exact inherited object, branch table, and common functional domain	Vacuum/scalar imported
W3.2	Geometric hypersurface-deformation Lie algebroid	History class
W3.3	Exact off-shell function algebra, on-shell conormal object, fixed-frame map	Vacuum/scalar algebra proved; no anchor claim
W3.4	Fixed-normal typed split regularity, characteristic equality, and WDW-2 intersection	Vacuum occupied; scalar conditional
W3.5	Coefficient-sensitive morphism and direct path/homotopy evaluation	Vacuum occupied; scalar conditional
W3.6	Nash–Moser evaluation equivalence, local pair-groupoid coherence, residuals	Vacuum occupied; scalar conditional
W3.7	Source, observable, boundary, relational, and clock coherence	Same passing branches
W3.8	Capstone closure and WDW-4 handoff	Branchwise

Proposition 49.1 (Acyclicity). *The W3.1–W3.8 graph is acyclic and no capstone conclusion consumes a conditional branch as though it were occupied.*

Proof. Every gate cites only earlier gates or an identified external import. The algebra gate ends with a fixed-frame bracket map and does not consume evaluation. Regularity does not consume a finite functor. W3.5 first proves differentiated evaluation and the complete coefficient-sensitive morphism and then defines the path functor directly. W3.6 consumes that functor and the W3.4 Green family to prove the local evaluation equivalence. Coherence consumes only constructed arrows. The branch qualifier is carried in every theorem and in the machine-readable dependency and claim-status ledgers. $\square$

50. Branchwise capstone theorem

Theorem W3.8 (Source-Fibre-Preserving, Boundary-Standing-Sensitive, Groupoid-Faithful Hypersurface-Deformation and Regular Refoliation Closure). Let b be a WDW-1/WDW-2-eligible branch and $z \in \mathcal{C}_{\text{tot}b}$. Assume the exact canonical, source, attachment, reconstruction, properness, and boundary certificates; the WDW-3 common-domain and loss-control conditions; a licensed reconstructed globally hyperbolic history; and, for conclusions requiring regularity, full split rank and absence of generalized KIDs at z . Then:

1. $\mathcal{E}_{\text{def}}$ is the geometric hypersurface-deformation Lie algebroid with metric-dependent bracket and embedding anchor.
2. For fixed descriptors, $C_b(\xi, N) = \mathcal{D}_b[\xi] + H_{\perp, b}[N]$ preserves that bracket through the Poisson bracket on $\mathcal{A}_{3b}$; on shell, $\mathcal{I}_b/\mathcal{I}_b^2 \simeq N^*\mathcal{C}_{\text{tot}b}$ supplies the $C^\infty(\mathcal{C}_{\text{tot}b})$ -linear constraint algebroid.
3. On the full regular locus, $\mathcal{C}_{\text{tot}b}$ is a split tame submanifold with compatible Sobolev-level split Hilbert realizations; on the occupied vacuum chart the splitting is supplied by $G_z = (A_z \mathcal{B}_{z,*} A_z^\#)^{-1}$ and $Q_z = \mathcal{B}_{z,*} A_z^\# G_z$, the same balanced pseudodifferential equation inverse on every nested level.
4. Its characteristic distribution is exactly

$$\ker(\Omega_b|_{\mathcal{C}_{\text{tot}b}}) = \{X_{C_b(\xi, N)} : (\xi, N) \in \mathcal{X} \oplus \mathcal{N}\}.$$

5. The accepted WDW-2 orbit/characteristic/vertical equality holds separately on $\mathcal{C}_{\text{sit}b}$, and its tangential distribution agrees with $\ker d\pi_{\text{prop}}$ and embeds in the full characteristic distribution at points of $\mathcal{C}_{\text{tot}b}$.
6. The augmented history-embedding groupoid $\mathbb{E}_b$ is explicitly constructed and its Lie algebroid is $\mathcal{E}_{\text{def}}$.

7. The guarded canonical characteristic path groupoid is explicitly constructed, and the direct path/homotopy evaluation functor has physical forward differential $d\text{Eval}(v_{\text{emb}}(\xi, N)) = X_{C_b(\xi, N)}$, with the conormal anchor square and coefficient-sensitive Leibniz identity carrying the declared right/fundamental signs.
8. On contractible regular embedding charts, direct and composite proper routes with common endpoints determine the same canonical arrow because no-KID injectivity, the typed descriptor inverse $e_z = -G_z A_z \mathcal{B}_{z,*} \mathcal{J}_{E,z} \Omega_z^b$, the resulting constructed characteristic projection, Nash–Moser local evaluation equivalence, and trivial plaque holonomy make the local path groupoids pair groupoids.
9. Globally, every residual is isotropy/KID, large or mapping-class action, boundary charge or flux, source/attachment change, relational-record change, target change, or failure of existence or regularity.
10. Proper arrows preserve

$$\hat{x}, \quad \text{Att}_{\hat{x}}, \quad W_{\hat{x}}, \quad \overline{\text{Recon}}_b, \quad \overline{\mathcal{O}}_b, \quad \mathcal{R}, \quad \text{BdyCert}_b,$$

and therefore cannot create distinct terminal standing merely by changing slicing.

11. KID points remain singular groupoid objects; charge-bearing boundary arrows, large transformations, and relationally distinct histories are not quotiented.
12. Lapse and foliation are presentation data only within the proper class; an eligible relational or boundary clock retains the standing assigned by its independent clock certificate.

All twelve conclusions are occupied on the explicit vacuum regular chart $\mathcal{U}_{\text{hyp}}$. Conclusions 1–2 are unconditionally proved for the exact massive Klein–Gordon lift; conclusions 3–12 apply there only at populated ScalarReg3 points. Internal and boundary extensions remain separately certificate-gated.

Proof. Conclusion 1 is W3.2. Conclusion 2 is W3.3. Conclusions 3–5 are W3.4 and the separate WDW-2 intersection theorem. Conclusions 6–7 are W3.5. Conclusions 8–9 are W3.6. Conclusion 10 is the source, observable, and clock coherence chain. Conclusion 11 is the singular and boundary non-collapse theorems. Conclusion 12 is the role classifier and clock compatibility theorem. Vacuum occupancy is example 25.1 and theorems 25.2 and 26.2; scalar and extension qualifiers are Certificates 20.1 and 27.1 and definition 41.1. Acyclicity and branch fidelity are proposition 49.1. $\square$

Corollary 50.1 (Complete classical vacuum closure). *The occupied vacuum chart satisfies the complete classical WDW deformation constraints, regular characteristic reduction, and local finite-refoliation closure on the declared common domain.*

Corollary 50.2 (Branchwise scalar closure). *The massive scalar lift satisfies exact off-shell HDA closure. Its full regular refoliation quotient is a theorem schema with a complete antecedent, not an occupied universal claim.*

51. WDW-4 handoff and conclusion

WDW-4 receives

$$\mathfrak{H}_{W3}(b) = \left(\mathcal{C}_{\text{tot}b} // \mathcal{C}_b^{\text{char}}, \mathcal{E}_{\text{def}b}, \mathfrak{A}_{\text{con},b}^{\text{reg}}, \text{Sing}_b^{\text{KID}}, \overline{\text{Src}}_b, \overline{\text{Recon}}_b, \overline{\mathcal{O}}_b, \mathcal{R}_b, \text{BdyCert}_b, \text{BranchStatus}_b \right).$$

The quotient notation denotes the constructed groupoid object with its regular local atlas and retained singular strata. It is not a global smooth manifold declaration.

The passing statuses are:

1. vacuum compact core: PROVED/OCCUPIED;

2. massive Klein–Gordon algebra: PROVED;
3. massive Klein–Gordon regular quotient: CONDITIONAL ON SCALARREG3;
4. nontrivial internal gauge: CONDITIONAL ON INTERNALCERT3;
5. finite/asymptotic boundary: CONDITIONAL ON BOUNDARY CERTIFICATE;
6. noncanonical forced source: BLOCKED;
7. quantum operator, anomaly, inner product, and quantum clock: OUT OF SCOPE.

No WDW-4 construction may silently promote a conditional branch. Quantization must act on the full algebroid/groupoid object, preserve its source and boundary coordinates, retain the singular strata or justify their quantum treatment, and state a regulator and operator domain. Classical closure does not by itself imply an anomaly-free quantum constraint representation.

Conventional result. The exact vacuum and scalar hypersurface-deformation brackets are proved off shell. The occupied vacuum branch has a split regular constraint locus, direct full characteristic equality, a conormal constraint algebroid, and a constructed local characteristic groupoid with exhaustive global residual guards.

AASC result. Once history, source, attachment, observable, relational, boundary, branch, and topology coordinates are fixed, a proper change of foliation leaves no standing-bearing residue. The quotient is forced by faithful same-target comparison, while every standing-bearing difference is retained.

ARC result. The final classical gate is closed for the occupied vacuum core. WDW-4 receives a branch-typed groupoid-level canonical object rather than a formal constraint equation or a privileged foliation.

Part X

Audit appendices

A. Complete convention ledger

Coordinate	Frozen value
Spacetime signature	$(-+++)$
Future normal	$n^\mu n_\mu = -1$, fixed by coorientation
Spatial metric	$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$
Extrinsic curvature	$K_{ij} = h_i^\mu h_j^\nu \nabla_\mu n_\nu = \frac{1}{2} \mathcal{L}_n h_{ij}$
Riemann tensor	$[\nabla_\mu, \nabla_\nu] v^\rho = R^\rho{}_{\sigma\mu\nu} v^\sigma$
Canonical momentum	$\pi^{ij} = \frac{\sqrt{h}}{2\kappa} (K^{ij} - K h^{ij})$, density weight one
Canonical two-form	$\Omega = \int \delta\pi^{ij} \wedge \delta h_{ij} + \delta p_\phi \wedge \delta\phi$
Hamiltonian convention	$\iota_{X_F} \Omega = -dF$
Poisson bracket	$\{F, G\} = dF(X_G)$
Spatial generator	$\mathcal{D}[\xi] = \int \pi^{ij} \mathcal{L}_\xi h_{ij} + p_\phi \mathcal{L}_\xi \phi$
Left-action moment map	$J_{\text{left}}(\xi) = -\mathcal{D}[\xi]$
Normal density	$\mathcal{H}_\perp^g = \frac{2\kappa}{\sqrt{h}} (\pi_{ij} \pi^{ij} - \frac{1}{2} \pi^2) - \frac{\sqrt{h}}{2\kappa} (R - 2\Lambda)$

Coordinate	Frozen value
Scalar potential	$V(\phi) = \frac{1}{2}m^2\phi^2$
Scalar momentum	$p_\phi = \sqrt{h} n^\mu \nabla_\mu \phi$, density weight one
Deformation bracket	$[(\xi, N), (\eta, M)]_h = ([\xi, \eta] + N \text{grad } M - M \text{grad } N, \mathcal{L}_\xi M - \mathcal{L}_\eta N)$
Physical embedding velocity	$v_{\text{emb}}(\xi, N) = i_*\xi + Nn$
Source algebroid anchor	$\rho_{\text{src}}(\xi, N) = -v_{\text{emb}}(\xi, N)$
Constraint Lie–Rinehart bracket	$[[F], [G]]_{\text{con}} = -\{F, G\}$ on $\mathcal{S}/\mathcal{S}^2$
Conormal anchor	$\rho_{\text{con}}([F]) = X_F _{\mathcal{C}_{\text{tot}}}$
Descriptor morphism	$\Psi(\xi, N) = -[C(\xi, N)]$
Normal–normal shift	$\beta_h(N, M)^i = h^{ij}(ND_j M - MD_j N)$
Boundary normal	Outward spacelike normal n_∂^i , distinct from spacetime normal n^μ
Core cosmological constant	$\Lambda = 0$ for the explicit occupied chart; formulas retain general Λ

B. Full vacuum functional bracket

This appendix exposes the cancellation classes behind equation (18). Put

$$\mathcal{P} = \pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2, \quad T[N] = \int \frac{2\kappa N}{\sqrt{h}} \mathcal{P}, \quad V[N] = - \int \frac{N\sqrt{h}}{2\kappa} (R - 2\Lambda).$$

At fixed π^{ij} ,

$$\delta\mathcal{P} = \left(2\pi^i_k \pi^{jk} - \pi\pi^{ij}\right) \delta h_{ij} + (2\pi_{ij} - \pi h_{ij}) \delta\pi^{ij}.$$

Also

$$\delta h^{-1/2} = -\frac{1}{2}h^{-1/2}h^{ij}\delta h_{ij}.$$

Therefore

$$\frac{\delta T[N]}{\delta\pi^{ij}} = \frac{4\kappa N}{\sqrt{h}} \left(\pi_{ij} - \frac{1}{2}\pi h_{ij}\right)$$

and

$$\frac{\delta T[N]}{\delta h_{ij}} = \frac{2\kappa N}{\sqrt{h}} \left[2\pi^i_k \pi^{jk} - \pi\pi^{ij} - \frac{1}{2}h^{ij}\mathcal{P}\right].$$

The curvature variation is

$$\delta(\sqrt{h}R) = \sqrt{h} \left[-G^{ij}\delta h_{ij} + D^i D^j \delta h_{ij} - D^2(h^{ij}\delta h_{ij})\right].$$

Multiplication by N and two integrations by parts yield

$$\frac{\delta V[N]}{\delta h_{ij}} = \frac{N\sqrt{h}}{2\kappa} \left(G^{ij} + \Lambda h^{ij}\right) - \frac{\sqrt{h}}{2\kappa} (D^i D^j N - h^{ij} D^2 N).$$

Thus

$$\{T[N], T[M]\} = 0$$

because both momentum and metric derivatives are ultralocal and proportional to the same smearing, while

$$\{V[N], V[M]\} = 0$$

because V has no momentum derivative. The non-derivative terms in $\{T[N], V[M]\} + \{V[N], T[M]\}$ are proportional to $NM - MN$. The surviving term is

$$-2 \int M(D^i D^j N - h^{ij} D^2 N) \left(\pi_{ij} - \frac{1}{2}\pi h_{ij}\right) - (N \leftrightarrow M).$$

Using

$$h^{ij} \left(\pi_{ij} - \frac{1}{2} \pi h_{ij} \right) = -\frac{1}{2} \pi$$

shows that the two trace contributions cancel, leaving

$$2 \int \pi^{ij} (N D_i D_j M - M D_i D_j N) = \int \pi^{ij} \mathcal{L}_{\beta_h(N, M)} h_{ij}.$$

This proves the off-shell equality without suppressing a cancellation.

C. Full scalar functional bracket

The scalar functional derivatives are

$$H_\phi[N]_{,p} = \frac{Np}{\sqrt{h}}, \quad H_\phi[N]_{,\phi} = -D_i(N\sqrt{h}D^i\phi) + N\sqrt{h}m^2\phi.$$

The bracket expands to

$$\begin{aligned} \{H_\phi[N], H_\phi[M]\} = & -\frac{Mp}{\sqrt{h}} [(D_i N)\sqrt{h}D^i\phi + N D_i(\sqrt{h}D^i\phi)] \\ & + \frac{Np}{\sqrt{h}} [(D_i M)\sqrt{h}D^i\phi + M D_i(\sqrt{h}D^i\phi)] \\ & + NMpm^2\phi - MNpm^2\phi. \end{aligned}$$

The second-derivative and mass rows cancel. The first row leaves

$$pD^i\phi(ND_iM - MD_iN) = p\beta^i D_i\phi.$$

For the metric cross bracket, write

$$\frac{\delta H_\phi[M]}{\delta h_{ij}} = M\sqrt{h} \left[\frac{1}{4} h^{ij} \left(-\frac{p^2}{h} + D_k\phi D^k\phi + m^2\phi^2 \right) - \frac{1}{2} D^i\phi D^j\phi \right]$$

in the fixed-density convention. This is MU^{ij} , while $\delta H_g[N]/\delta\pi^{ij} = NA_{ij}$. The antisymmetrized cross term is exactly zero.

D. Lie-algebroid Jacobi calculation

For $e_a = (\xi_a, N_a)$, set

$$[e_a, e_b]_h = (\Xi_{ab}, \nu_{ab}),$$

$$\Xi_{ab} = [\xi_a, \xi_b] + N_a \text{grad } N_b - N_b \text{grad } N_a, \quad \nu_{ab} = \mathcal{L}_{\xi_a} N_b - \mathcal{L}_{\xi_b} N_a.$$

The apparent fixed- h Jacobiator contains terms such as

$$N_1 \text{grad}(\langle \xi_2, dN_3 \rangle - \langle \xi_3, dN_2 \rangle)$$

and terms with two gradients. The source anchor differentiates the inverse metric:

$$\rho_{\text{src}}(e_1)(h^{ij}) = -h^{ik}h^{j\ell}\rho_{\text{src}}(e_1)(h_{k\ell}).$$

For the frozen right/fundamental convention,

$$\boxed{\rho_{\text{src}}(\xi_1, N_1)(h_{ij}) = -\mathcal{L}_{\xi_1} h_{ij} - 2N_1 K_{ij}, \quad \rho_{\text{src}}(\xi_1, N_1)(h^{ij}) = -\mathcal{L}_{\xi_1} h^{ij} + 2N_1 K^{ij}.}$$

This is also the coefficient derivative in the Hamiltonian audit, because

$$\{C(e), f(h)\} = -X_{C(e)}f = \rho_{\text{src}}(e)f.$$

For a fully displayed normal–normal–normal sector put $e_a = (0, N_a)$ and

$$\beta_{bc}^i = h^{ij}(N_b D_j N_c - N_c D_j N_b).$$

The source-anchor variation of the field-dependent coefficient is

$$\rho_{\text{src}}(e_a)\beta_{bc}^i = 2N_a K^{ij}(N_b D_j N_c - N_c D_j N_b). \quad (61)$$

Its cyclic sum vanishes pairwise:

$$\sum_{\text{cyc}} \rho_{\text{src}}(e_a)\beta_{bc}^i = 0.$$

The normal component of the nested bracket is

$$-\sum_{\text{cyc}} \beta_{bc}^j D_j N_a = -\sum_{\text{cyc}} h^{jk}(N_b D_k N_c - N_c D_k N_b) D_j N_a = 0,$$

by pairwise cancellation and symmetry of h^{jk} . The change of the Gaussian extension contributes

$$\mathcal{E}_{a|bc}^i = -2N_a K^{ij}(N_b D_j N_c - N_c D_j N_b);$$

it is the negative of equation (61). Thus the raw-extension correction and source-anchor correction use the same frozen sign before cyclic summation. The complete audited identity is

$$\sum_{\text{cyc}} \left([e_a, [e_b, e_c]_h]_h + \delta_{\rho_{\text{src}}(e_a)}[e_b, e_c]_h \right) = 0. \quad (62)$$

For general lapse–shift descriptors, the Lie-derivative part converts ordinary derivatives into tensorial commutators, while the normal part is organized exactly as above. After cyclic summation:

1. the pure spatial terms vanish by vector-field Jacobi;
2. the one-normal terms vanish by $[\mathcal{L}_{\xi_1}, \mathcal{L}_{\xi_2}] = \mathcal{L}_{[\xi_1, \xi_2]}$;
3. the two-normal terms cancel against $\rho(h^{-1})$ contributions; and
4. the three-normal scalar component vanishes by antisymmetry of $N_a D_i N_b - N_b D_i N_a$.

This is the component image of the zero Jacobiator of the ambient Gaussian vector-field commutator. Freezing h removes item 3 and produces the known false Lie-algebra calculation.

E. Sobolev scale and derivative-loss ledger

Map	Source	Target	Control
Metric inversion, determinant	$h \in H^s, s > \frac{3}{2}$	H^s	Smooth tame on positive metrics
Scalar curvature	$h \in H^s$	H^{s-2}	Two-derivative loss
Momentum divergence	$(h, \pi) \in H^s \times H^{s-1}$	H^{s-2}	One derivative of π , one connection factor
Scalar energy	$(h, \phi, p) \in H^s \times H^s \times H^{s-1}$	H^{s-2}	Sobolev multiplication
Spatial Lie derivative	$(q, \xi) \in H^{s+1} \times H^{s+1}$	H^s	One-level transport loss
Normal Hamiltonian field	Smooth inverse limit	Smooth inverse limit	No fixed-index global flow claim
$D\Phi$	$E_r = H^{r+2}(S^2) \oplus H^{r+1}(S^2)$	$F_r = H^r(\text{Dens}) \oplus H^r(T^* \otimes \text{Dens})$	Orders $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$

Map	Source	Target	Control
Formal adjoint $A^\#$	$\widehat{Z}_r = H^{r+2} \oplus H^{r+2}(T)$	$\widetilde{E}_r = H^r(S^2) \oplus H^{r+1}(S^2)$	Differential formal adjoint; distinct from A' and $A^\dagger$
$\mathcal{B}_{z,*}$	$\widetilde{E}_r$	E_r	Fixed positive row balancer of component orders $(-2, 0)$
Banach transpose A'	F'_r	E'_r	Distributional factorization
Hilbert adjoint $A^\dagger$ $L = A\mathcal{B}_*A^\#$	F_r , after Riesz map $\widehat{Z}_r$	E_r , after Riesz map F_r	Order-reduced realization One fixed order-two elliptic, self-adjoint, index-zero pseudodifferential operator
$G = L^{-1}$	F_r	$\widehat{Z}_r$	Unique compatible multiplier inverse; coefficient loss $\mu = 2$
$Q = \mathcal{B}_*A^\#G$	F_r	E_r	Compatible tame right inverse; $AQ = 1$
Eval	$\mathcal{E}_r = \text{Emb}^{r+3}$	$\mathcal{P}_r = H^{r+2} \oplus H^{r+1}$	Smooth tame; fixed loss $\ell = 2$
$R_i = (d\text{Eval}_i)^{-1}$	$\mathcal{K}_z \cap \mathcal{P}_{r+3}$	$T_i\mathcal{O}_{\text{emb}} \cap \mathcal{E}_r$	$-v_{\text{emb}}GAB_*\mathcal{J}_E\Omega^b$; $\ell_R = 3$, $\mu = 2$
Guarded path evaluation	$C_t^0\mathcal{E}_{r+1} \cap C_t^1\mathcal{E}_r$	$C_t^0\mathcal{P}_{r+1} \cap C_t^1\mathcal{P}_r$	Direct on paths and homotopies
Local Cauchy evolution	Gauge-fixed hyperbolic data	Local spacetime solution	Separate PDE theorem, not Fréchet ODE
Reduced slice form	$T^*S_a^r$	Strong Hilbert symplectic form	Local completion only

F. Linearized constraint and KID operators

The principal part of the vacuum linearized constraint map is

$$D\Phi_g(\gamma, k)_{\text{prin}} = \left(-D^i D^j \gamma_{ij} + D^2 \text{tr } \gamma, D_j k^j_i - D_i \text{tr } k \right).$$

Its formal adjoint principal part is

$$D\Phi_g^*(N, X)_{\text{prin}} = \left(D_i D_j N - h_{ij} D^2 N, -D_{(i} X_{j)} + h_{ij} D_k X^k \right),$$

up to the invertible bundle maps produced by the Legendre convention. The lower-order terms restore the exact KID system equations (32) and (33).

For the scalar branch, the linearized Hamiltonian row additionally contains

$$\delta\mathcal{H}_\perp^\phi = \frac{p_\phi}{\sqrt{h}} \delta p_\phi + \sqrt{h} D^i \phi D_i \delta\phi + \sqrt{h} m^2 \phi \delta\phi + \text{metric terms},$$

and the momentum row contains

$$\delta\mathcal{H}_{i_i}^\phi = \delta p_\phi D_i \phi + p_\phi D_i \delta\phi.$$

The adjoint kernel therefore includes preservation of both the metric and scalar history. No vacuum adjoint formula is reused as a scalar certificate.

G. Occupied vacuum regular chart

The occupied data are summarized by

Coordinate	Value
Σ	$\mathbb{H}^3/\Gamma$, Γ torsion-free cocompact
h	sectional curvature -1
Λ	0
K_{ij}	h_{ij}
π^{ij}	$-(\sqrt{h}/\kappa)h^{ij}$
R	-6
K	3
$\mathcal{H}_\perp$	0
$\mathcal{H}_{ii}$	0
KID	0

The constraint equalities and KID proof are in example 25.1 and theorem 25.2. The normal Green operator and right inverse are

$$G_z = (A_z \mathcal{B}_{z,*} A_z^\#)^{-1} : F_r \rightarrow \widehat{Z}_r, \quad Q_z = \mathcal{B}_{z,*} A_z^\# G_z : F_r \rightarrow E_r.$$

The raw differential expression $A_z A_z^\#$ retains its unequal row weights and is not inverted on F_r . The row balancer is part of one Sobolev-index-independent pseudodifferential equation; uniqueness plus elliptic regularity therefore proves compatibility. The open regular chart is chosen small enough that the inverse persists, the WDW-2 spatial slice remains valid, and evaluation lies in the local globally hyperbolic development class.

H. Scalar regularity certificate

The status of the six ScalarReg3 coordinates is:

Coordinate	Theorem status
Exact coupled constraint map	PROVED
Exact off-shell total HDA	PROVED
One explicit coupled constraint point	NOT USED AS REGULAR OCCUPANT
Zero generalized-KID adjoint kernel at that point	CONDITIONAL
Closed range and tame right inverse	CONDITIONAL
Licensed pointed Einstein–Klein–Gordon history	CONDITIONAL

Accordingly, the manuscript makes no unqualified scalar regular-quotient assertion.

I. History-embedding groupoid structure maps

For

$$\alpha = (a_1, i_1; a_0, i_0; \gamma; \chi),$$

the complete map ledger is:

$$\begin{aligned} s(\alpha) &= (a_0, \gamma, i_0, \text{Eval}_{i_0} \gamma, \hat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R}), \\ t(\alpha) &= (a_1, \gamma, i_1, \text{Eval}_{i_1} \gamma, \hat{x}, \text{Att}, W, \text{BdyCert}, \mathcal{R}), \\ u(a, \gamma, i, \dots) &= (a, i; a, i; \gamma; \text{id}), \\ \alpha^{-1} &= (a_0, i_0; a_1, i_1; \gamma; \chi^{-1}), \\ \alpha_{21} \circ \alpha_{10} &= (a_2, i_2; a_0, i_0; \gamma; \chi_{21} \circ \chi_{10}). \end{aligned}$$

Restriction acts simultaneously on the embedding chart and presentation cover. Augmentation changes only the cover coordinate. Isotropy at i consists of proper history isometries fixing the evaluated object; at a KID point its Lie algebra is the KID space.

On a full-regular chart U , the canonical target is independently constructed as the guarded characteristic path groupoid $\mathbb{C}_{b,U}^{\text{char}} \rightrightarrows U_{\text{aug}}$. For a characteristic path c ,

$$s[c] = c(0), \quad t[c] = c(1), \quad 1_z = [\text{const}_z], \quad [c]^{-1} = [c^{-1}], \quad [c_2] \circ [c_1] = [c_2 * c_1].$$

Path reversal and concatenation belong to this target path construction; the source history groupoid continues to use χ^{-1} and $\chi_{21} \circ \chi_{10}$. The direct map is

$$\text{Eval}^{\text{path}} : \Pi_1^{\text{guard}}(\mathcal{O}_{\text{emb}}) \longrightarrow \mathbb{C}_{b,U}^{\text{char}}.$$

Only after the no-KID Nash–Moser local-equivalence and trivial-holonomy theorem are proved do both guarded path groupoids reduce locally to pair groupoids and the functor descend to the χ -arrow presentation.

J. Evaluation and reconstruction coherence diagrams

The local monodromy square is

$$\begin{array}{ccc} (\gamma, i_0, \hat{x}, \text{Att}, W) & \xrightarrow{[i_t]} & (\gamma, i_1, \hat{x}, \text{Att}, W) \\ \text{Eval}_{i_0} \downarrow & & \downarrow \text{Eval}_{i_1} \\ z_0 & \xrightarrow{[\text{Eval}_{i_t} \gamma]} & z_1. \end{array}$$

The source square is

$$\begin{array}{ccc} (\gamma, i_0) & \xrightarrow{[i_t]} & (\gamma, i_1) \\ \overline{\text{SrcEval}}_{i_0} \downarrow & & \downarrow \overline{\text{SrcEval}}_{i_1} \\ \hat{x} & \xrightarrow{\text{id}} & \hat{x}, \end{array}$$

and the reconstruction triangle is

$$\overline{\text{Recon}}([z_0]) \simeq \gamma \simeq \overline{\text{Recon}}([z_1]).$$

The same diagrams carry $\text{Att}, W, \text{BdyCert}, \mathcal{R}$. No arrow is inferred from tensor equality. On the occupied regular chart, the evaluation local-diffeomorphism and trivial-holonomy theorem turn these path-class arrows into unique endpoint arrows.

K. Boundary generator ledger

Class	Variation status	Improvement	Charge test	Quotient status
Compact core	No surface term	None	Identically zero	Proper/occupied
Finite Dirichlet	Boundary term fixed by induced geometry	Ensemble-specific	Must be integrable and zero	Conditional
Finite mixed	Flux generally present	Ensemble-specific	Flux and charge	Conditional
Asymptotically flat translation	RT falloff/parity	ADM surface term	E_{ADM}, P_i	Nonzero charge retained
Asymptotic rotation/boost	RT falloff/parity	Angular/boost charge	J_{ij}, K_i	Nonzero charge retained
Large transformation	Differentiability branch-specific	Global improvement	Sector/mapping class	Retained

L. HKT assumption matrix

Source assumption	WDW-3 match	Consequence
Metric and conjugate momentum are sole variables	Vacuum branch only	Scalar/internal branches excluded from the uniqueness comparison
Local differential generator	WDW-1 vacuum density	Agreement on overlapping functional class
Hypersurface-deformation relations	Directly proved in WDW-3	Consistency input for comparison, not an unproved premise
Time reversibility	Vacuum bulk overlap	No claim for irreversible enriched canonical laws
Boundaryless setting	Occupied core	No boundary uniqueness promotion
Specified derivative/order hypotheses	Exact HKT class only	No higher-derivative or nonlocal promotion

M. Adversarial map

Attack	Closure location
The Dirac algebra was merely quoted.	Sections 16–19 and Appendices B–C.
The self-bracket holds only on shell.	No constraint is used in equations (18) and (22).
Structure functions were treated as constants.	Sections 13 and 22; Appendix D.
A formal bracket was confused with finite refoliation.	The target is constructed before the direct path functor; Sections 30–38 derive the local pair-groupoid form.
The embedding groupoid does not act on ADM data.	The on-shell morphism Ψ , direct $\text{Eval}^{\text{path}}$, and equation (49).
The BFW phase-space gap was ignored.	Section 15 identifies the gap; W3.3 constructs the conormal algebroid, W3.5 proves the coefficient-sensitive morphism and path functor, and W3.6 supplies the typed fixed-normal inverse.

Attack	Closure location
One local slice became a global selector.	WDW-2 action groupoid retained; Section 45.
A fixed lapse was treated as basic.	Lemma 8.1; full $\mathcal{N}$ retained.
Normal flow is undefined on the claimed Hilbert carrier.	Smooth tower and explicit no-fixed-index-flow non-claim.
The total constraint surface is not always a manifold.	The fixed row-balanced $A_z\mathcal{B}_{z,*}A_z^\#$ inverse proves the typed equal-target split theorem only on the no-KID chart; KID strata remain singular.
Generic no-KID data do not prove occupancy.	Explicit hyperbolic occupant in Section 25.
Scalar regularity was imported from a corrected theorem.	ScalarReg3 gate and erratum.
KID points were deleted.	Section 29 and singular groupoid objects.
The Hamiltonian constraint was assumed.	WDW-1 derivation plus WDW-3 direct closure.
Refoliation invariance was interpretive.	Constructed target, direct path evaluation, typed multiplier inverse, constructed characteristic projection, Nash–Moser equivalence, holonomy, source, and observable theorems.
All spacetime diffeomorphisms were called gauge.	Section 42 projectability audit.
Boundary time was erased.	Sections 40–41 and Appendix K.
Relational change was mistaken for gauge stasis.	Sections 37 and 46.
A smooth Cauchy splitting selected a foliation.	Section 45 existence/selection theorem.
HKT was circular.	Section 47 assumption matrix and dependency order.
AASC replaced the GR calculation.	Conventional Parts III–VII precede Part VIII.
The kernel was treated as optional.	Section 43 dependency statement.
Source identity was inferred from a tensor.	Structured coordinates and lemma 9.1.
Large transformations were collapsed.	Global residual classes 2–4.
The quantum problem of time was solved.	Explicit WDW-4 nonclaims and handoff.

N. Machine-readable audit schemas

The ‘audit/’ directory contains the following rectangular UTF-8 CSV ledgers:

1. WDW-1 and WDW-2 import ledgers;
2. external theorem imports;
3. branch eligibility and canonical conventions;
4. functional domain and Sobolev loss;
5. bracket terms and structure-function Jacobi;
6. deformation algebroid and total constraint map;
7. KID strata, vacuum occupancy, and scalar regularity;
8. characteristic equality;
9. embedding groupoid and evaluation/reconstruction;

10. finite routes and global residual classifier;
11. source, observable, boundary, projectability, and clock ledgers;
12. HKT assumptions, theorem dependency, claim status, adversarial map;
13. WDW-4 handoff and acceptance conditions.

The only permitted status values are

{PROVED, IMPORTED VERIFIED, OCCUPIED, CONDITIONAL, BLOCKED, OUT OF SCOPE}.

Every row has a branch, scope, evidence, and downstream-use field where applicable. Publication consistency requires agreement among theorem identifiers, branch predicates, status values, residual-class numbering, source/target groupoid operations, and downstream dependencies. Tabular agreement alone is not treated as semantic agreement.

O. Publication acceptance contract

The publication standard contains the following forty-eight conditions. The manuscript passes WDW-3 only when every applicable condition passes.

1. The final WDW-1 and WDW-2 publication interfaces are identified by theorem and scope.
2. Every imported theorem has a source theorem, hypotheses, specialization, and non-overclaim entry.
3. The compact core and all conditional extensions are separated.
4. Every conclusion is branch-indexed.
5. The full lapse family is retained; no fixed lapse is substituted.
6. The smooth carrier and mixed Sobolev scale are explicitly typed.
7. Every normal-deformation map has a derivative-loss ledger.
8. No global fixed-index Hilbert action or flow is claimed.
9. The common functional domain is stable under every bracket used.
10. The spatial–spatial bracket matches WDW-2 signs.
11. The spatial–normal bracket matches WDW-2 signs.
12. The vacuum normal–normal bracket is calculated directly off shell.
13. The scalar normal–normal and cross brackets are calculated directly off shell.
14. Field-dependent structure functions are handled in the Jacobi audit.
15. The geometric deformation algebroid is explicitly constructed.
16. Signature and anchor conventions are aligned with the canonical representation.
17. The Blohmann–Fernandes–Weinstein phase-space descent gap is acknowledged and not imported as solved.
18. The actual WDW-3 evaluation functor is constructed.
19. The total constraint map and formal adjoint are explicitly typed.
20. KIDs are identified with the correct adjoint kernel under stated hypotheses.
21. One occupied vacuum no-KID solution is produced.
22. Genericity is not used as a substitute for occupancy.

23. Massive scalar regularity is not promoted without ScalarReg3.
24. The regular zero locus is proved split by a valid tame/ILH argument.
25. Characteristic equality is proved directly without a weak double-annihilator shortcut.
26. The accepted WDW-2 spatial equality is recovered as a restriction.
27. KID points and isotropy remain in the global object.
28. The history-embedding groupoid has all structure maps and guard proofs.
29. Its Lie algebroid matches the geometric deformation bracket.
30. Differentiated evaluation matches the canonical constraint representation.
31. Local finite path coherence is proved on a declared chart.
32. Global residuals are exhaustively classified rather than assumed absent.
33. Source fibre, attachment, witness, and reconstruction ancestry are preserved.
34. Dirac, relational, record, and boundary observables are distinguished.
35. Boundary generators are varied and improved before charges are classified.
36. No charge-bearing transformation is quotiented as proper refoliation.
37. Projectable canonical descriptors are distinguished from arbitrary spacetime vector fields.
38. Lapse and foliation are not called physical clocks without the clock certificate.
39. No preferred foliation is introduced through a local slice or numerical convenience.
40. The HKT comparison lists all assumptions and is not circular.
41. The AASC kernel remains logically upstream and is not generated by refoliation success.
42. W3.1–W3.8 form one acyclic dependency chain.
43. WDW-4 receives only passing branches.
44. The adversarial map is checked before any verdict.
45. The manuscript, ledgers, abstract, conclusion, and capstone report the same endpoint.
46. The PDF contains no unresolved references, clipped mathematics, or unreadable tables.
47. All fonts are embedded and the PDF passes integrity testing.
48. A clean source-only build reproduces the publication PDF.

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